Resultats des tests UC - pili-pos v1.0.0

Date : 2026-09-25 **Reference :** §IV du manuel de certification (Procedure de communication GraphQL).

Ce document parcourt les UC obligatoires du Basic Set v1.8 (22 UC spec) + les 5 UC Reports v1.3 + les cinq situations d'arrondi du manuel (ROUND1 a ROUND5). Aucun UC n'est hors périmètre : **M122** (changement de centre de coût, transfert de table) et **M132** (correction de paiement) sont rejoués, voir leur section ; **M112A** n'est pas un parcours proposé par pili-pos (le remboursement partiel est toujours émis en ticket séparé, parcours M112B). La facture (**M150**, événement I) est rejouée comme les autres UC ; les deux UC de facture de l'Advanced Set (**UC4121** et **UC4122**) le sont aussi, dans la section qui suit les rapports. Pour chaque UC, il fournit :

- Le scenario d'execution (extrait de la fixture officielle SPF Finances).
- La requete et la reponse de reference de la fixture (fichiers Variables.json et *_Response.json publies).
- Le tableau "Champs a verifier" de la fixture.
- La reference au test automatise qui valide la conformite.
- Pour chaque UC rejoue : la requete **construite par pili-pos** et sa comparaison avec la fixture (bloc « Preuve pili-pos »).
- Les tickets correspondants sont transcrits dans *TICKETS_EXEMPLES_FR.md*.

Note importante : les fixtures SPF Finances etant la source de verite, les tableaux et JSON de reference ci-dessous correspondent exactement a ceux publies par le SPF dans `docs/20260821_UC_BasicSet_POS_FDM_1.8/` et `docs/20260821_UC_Reports_1.3/`. Les tests `crates/pos-fiscal/tests/uc_fixtures.rs` et `reports_fixtures.rs` verifient que pili-pos lit chaque fixture ; les rejeux applicatifs `crates/pos-engine/tests/spf_e2e_fixtures.rs` et `spf_e2e_report_fixtures.rs` verifient que pili-pos **produit** les memes requetes. Tous tournent a chaque commit (CI).

Revision des jeux de tests : Basic Set v1.8 (changelog : M121, reponse sans espace final dans `terminalId`), Advanced Set v1.6 (UC226, UC243, UC4119bis_2, UC4121 corriges, voir *COUVERTURE_ADVANCED_SET_FR.md*) et Reports v1.3 republie ; tous les horodatages des fixtures portent les millisecondes. Les JSON de reference ci-dessous sont ceux de ces dossiers.

Preuves generees par l'implementation : chaque UC rejoue se termine par un bloc « Preuve pili-pos (build `cert`) ». Le rejeu construit la commande a partir des produits, quantites et paiements de la fixture, la fait passer par les vraies commandes du binaire de certification (`--no-default-features --features cert`) et compare champ par champ la requete archivee a celle de la fixture : champs metier egaux, champs d'identification (posId, deviceId, numero de ticket, dates, periode de booking) controles sur leur format, ecarts assumes listes et motives. Le bloc reprend la requete **reellement construite par pili-pos** (colonne `event_data_json` de l'archive fiscale) et le resultat de la comparaison ; les fichiers sont dans `evidence/uc/` (`<UC>_event<N>_<label>_request.json`, `_response.json`, `<UC>_check.json`). Regeneration : `scripts/capture_uc_evidence.sh`. Le FDM y est un mock qui rejoue la reponse de la fixture ; une journee d'exploitation est signee par le simulateur en ligne module2 lors de la repetition (*JOURNAL_TESTS_PRE_SOUMISSION_FR.md*).

Note, terminal de paiement relie : les resultats ci-dessous sont ceux d'une caisse sans terminal de paiement relie (carte saisie par l'operateur, ligne de paiement `MANUAL`). Les lignes de paiement `AUTOMATIC` d'un terminal relie (type de carte fourni par le terminal, marque dans `provider`, identifiant de transaction dans `reference`) ne figurent pas dans les fixtures du SPF ; elles sont couvertes par les tests du moteur `crates/pos-engine/tests/eft_payments.rs` et `crates/pos-engine/tests/eft_signing.rs`. Aucun resultat enregistre ci-dessous n'est modifie.

Basic Set v1.8 (22 UC)

M110_signSale

Mutation invoquee : `signSale`

Scenario :

```
DINING ROOM:
Direct Sale of
    - 2 Dry Martini (unit price 12 EUR, VAT code A)
    - 1 Burger of the Chef (unit price 28 EUR, VAT code B).
Payment by CASH (52 EUR)
```

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	1000	OK
posDateTime	2024-07-29T13:00:00.000+02:00	OK
terminalId	TER-2-DIN	OK
deviceId	1631678d-7a85-4ac3-b296-bb4565e873fe	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	PAPER	OK
employeeId	84022899837	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	N	OK
fdmRef.eventCounter	15	OK
fdmRef.totalCounter	59	OK
shortSignature (presence)	40 caracteres hex	OK
vatCalc[] (calcul TVA par taux)	2 taux	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1000,
    "posDateTime": "2024-07-29T13:00:00.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "1631678d-7a85-4ac3-b296-bb4565e873fe",
    "terminalId": "TER-2-DIN",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "PAPER",
    "employeeId": "84022899837",
    "transaction": {
      "transactionLines": [
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "10006",
            "productName": "Dry Martini",
            "departmentId": "10",
```

```
            "departmentName": "Aperitifs",
            "quantity": 2,
            "quantityType": "PIECE",
            "unitPrice": 12,
            "vats": [
              {
                "label": "A",
                "price": 24
              }
            ]
          },
          "lineTotal": 24
        },
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "22001",
            "productName": "Burger of the Chef",
            "departmentId": "22",
            "departmentName": "Main Dishes",
            "quantity": 1,
            "quantityType": "PIECE",
            "unitPrice": 28,
            "vats": [
              {
                "label": "B",
                "price": 28
              }
            ]
          },
          "lineTotal": 28
        }
      ],
      "transactionTotal": 52
    },
    "financials": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "inputMethod": "MANUAL",
        "amount": 52,
        "amountType": "PAYMENT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
```

```
    "signSale": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1000,
      "posDateTime": "2024-07-29T13:00:00.000+02:00",
      "terminalId": "TER-2-DIN",
      "deviceId": "1631678d-7a85-4ac3-b296-bb4565e873fe",
      "vatCalc": [
        {
          "label": "A",
          "rate": 21,
          "taxableAmount": 19.83,
          "vatAmount": 4.17,
          "totalAmount": 24.0,
          "outOfScope": false
        },
        {
          "label": "B",
          "rate": 12,
          "taxableAmount": 25.0,
          "vatAmount": 3.0,
          "totalAmount": 28.0,
          "outOfScope": false
        }
      ],
      "eventOperation": "SALE",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T11:00:08.000Z",
        "eventLabel": "N",
        "eventCounter": 15,
        "totalCounter": 59
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "shortSignature": "ca39a3ee5e6b4b0d3255bfef95601890afd80709",
      "verificationUrl": "https://gs2.be/",
      "warnings": [],
      "informations": [],
      "footer": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m110_sale` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m110_sale`. Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M110_event1_N_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "e3f9c1af-8907-4d24-aaf1-c95f77ff25f0",
  "costCenter": {
    "id": "T01",
```

```
  "reference": "01",
  "type": "TABLE"
},
"deviceId": "F0:1D:3D:CC:13:C8",
"employeeId": "84022899837",
"estNo": "2000000042",
"financials": [
  {
    "amount": 52.0,
    "amountType": "PAYMENT",
    "id": "1",
    "inputMethod": "MANUAL",
    "name": "Cash",
    "type": "CASH"
  }
],
"language": "FR",
"posDateTime": "2026-10-18T06:03:47.808+02:00",
"posFiscalTicketNo": 1,
"posId": "CPOS0031234567",
"posSwVersion": "pili-pos/1.0.0",
"terminalId": "POS-1-BAR",
"ticketMedium": "PAPER",
"transaction": {
  "transactionLines": [
    {
      "lineTotal": 24.0,
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "departmentId": "10",
        "departmentName": "Aperitifs",
        "productId": "10006",
        "productName": "Dry Martini",
        "quantity": 2.0,
        "quantityType": "PIECE",
        "unitPrice": 12.0,
        "vats": [
          {
            "label": "A",
            "price": 24.0
          }
        ]
      }
    },
    {
      "lineTotal": 28.0,
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "departmentId": "22",
        "departmentName": "Main Dishes",
        "productId": "22001",
        "productName": "Burger of the CHEF",
        "quantity": 1.0,
        "quantityType": "PIECE",
        "unitPrice": 28.0,
```

```
          "vats": [
            {
              "label": "B",
              "price": 28.0
            }
          ]
        }
      }
    ],
    "transactionTotal": 52.0
  },
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M110_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Écarts assumés :

Champ	Fixture	pili-pos	Motif
`costCenter.type`	-	TABLE	une vente en salle est encaissée depuis sa table et porte le centre de coût TABLE

M111_signSale_Refund

Mutation invoquee : `signSale`

Scenario :

```
Dining room:
Refund ticket (full refund):
Products on original ticket N 1000/1200:
    - 2 Dry Martini (unit price 12 EUR, VAT code A)
    - 1 Burger of the Chef (unit price 28 EUR, VAT code B)
```

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	NL	OK
`posId`	CPOS0031234567	OK
`posFiscalTicketNo`	1001	OK
`posDateTime`	2024-07-29T13:15:00.000+02:00	OK
`terminalId`	TER-2-DIN	OK
`deviceId`	1631678d-7a85-4ac3-b296-bb4565e873fe	OK
`vatNo`	BE0000000097	OK
`estNo`	2000000042	OK
`posSwVersion`	1.8.3	OK
`bookingPeriodId`	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
`bookingDate`	2024-07-29	OK
`ticketMedium`	PAPER	OK
`employeeId`	84022899837	OK
`fdmRef.fdmId`	FOD01000001	OK
`fdmRef.eventLabel`	N	OK

Champ	Valeur fixture	Statut
fdmRef.eventCounter	16	OK
fdmRef.totalCounter	60	OK
shortSignature (presence)	40 caracteres hex	OK
vatCalc[] (calcul TVA par taux)	2 taux	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1001,
    "posDateTime": "2024-07-29T13:15:00.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "1631678d-7a85-4ac3-b296-bb4565e873fe",
    "terminalId": "TER-2-DIN",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "PAPER",
    "employeeId": "84022899837",
    "fdmRef": {
      "fdmId": "FOD01000001",
      "fdmDateTime": "2024-07-29T11:00:08.000Z",
      "eventLabel": "N",
      "eventCounter": 15,
      "totalCounter": 59
    },
    "transaction": {
      "transactionLines": [
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "10006",
            "productName": "Dry Martini",
            "departmentId": "10",
            "departmentName": "Aperitifs",
            "quantity": -2,
            "quantityType": "PIECE",
            "negQuantityReason": "REFUND",
            "unitPrice": 12,
            "vats": [
              {
                "label": "A",
                "price": -24
              }
            ]
          },
          "lineTotal": -24
        },
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
```

```
            "productId": "22001",
            "productName": "Burger of the Chef",
            "departmentId": "22",
            "departmentName": "Main Dishes",
            "quantity": -1,
            "quantityType": "PIECE",
            "negQuantityReason": "REFUND",
            "unitPrice": 28,
            "vats": [
              {
                "label": "B",
                "price": -28
              }
            ]
          },
          "lineTotal": -28
        }
      ],
      "transactionTotal": -52
    },
    "financials": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "inputMethod": "MANUAL",
        "amount": -52,
        "amountType": "PAYMENT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signSale": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1001,
      "posDateTime": "2024-07-29T13:15:00.000+02:00",
      "terminalId": "TER-2-DIN",
      "deviceId": "1631678d-7a85-4ac3-b296-bb4565e873fe",
      "vatCalc": [
        {
          "label": "A",
          "rate": 21,
          "taxableAmount": -19.83,
          "vatAmount": -4.17,
          "totalAmount": -24.0,
          "outOfScope": false
        },
```

```
        {
          "label": "B",
          "rate": 12,
          "taxableAmount": -25.0,
          "vatAmount": -3.0,
          "totalAmount": -28.0,
          "outOfScope": false
        }
      ],
      "eventOperation": "SALE",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T11:15:00.000Z",
        "eventLabel": "N",
        "eventCounter": 16,
        "totalCounter": 60
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "shortSignature": "ca39a3ee5e6b4b0d3255bfef95601890afd80709",
      "verificationUrl": "https://gs2.be/",
      "warnings": [],
      "informations": null
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m111_full_refund` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m111_full_re`
Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M111_event2_N_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "8eef7f3e-8f1f-4b9a-92b3-aefca5a66603",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "84022899837",
  "estNo": "2000000042",
  "fdmRef": {
    "eventCounter": 15,
    "eventLabel": "N",
    "fdmDateTime": "2024-07-29T11:00:08.000Z",
    "fdmId": "FOD01000001",
    "totalCounter": 59
  },
  "financials": [
    {
      "amount": -52.0,
      "amountType": "PAYMENT",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
```

```
  }
],
"language": "FR",
"posDateTime": "2026-10-18T06:03:47.945+02:00",
"posFiscalTicketNo": 2,
"posId": "CPOS0031234567",
"posSwVersion": "pili-pos/1.0.0",
"terminalId": "POS-1-BAR",
"ticketMedium": "PAPER",
"transaction": {
  "transactionLines": [
    {
      "lineTotal": -24.0,
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "departmentId": "10",
        "departmentName": "Aperitifs",
        "negQuantityReason": "REFUND",
        "productId": "10006",
        "productName": "Dry Martini",
        "quantity": -2.0,
        "quantityType": "PIECE",
        "unitPrice": 12.0,
        "vats": [
          {
            "label": "A",
            "price": -24.0
          }
        ]
      }
    },
    {
      "lineTotal": -28.0,
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "departmentId": "22",
        "departmentName": "Main Dishes",
        "negQuantityReason": "REFUND",
        "productId": "22001",
        "productName": "Burger of the CHEF",
        "quantity": -1.0,
        "quantityType": "PIECE",
        "unitPrice": 28.0,
        "vats": [
          {
            "label": "B",
            "price": -28.0
          }
        ]
      }
    }
  ],
  "transactionTotal": -52.0
},
"vatNo": "BE0000000097"
```

```
}
```

Events préalables du scénario, archivés de même : `evidence/uc/M111_event1_N_request.json`. Réponses FDM archivées : fichiers `*_response.json` correspondants.

Comparaison avec la fixture (`evidence/uc/M111_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M112A_signSale_RefundPartial

Scenario global :

```
scenario M112A:
ASSUMPTION: the cash register supports a partial refund in combination with a new sale

1. Original Ticket: direct sale of
    * 2 dry Martini (unit price 12 EUR, VAT code A)
    * 1 burger of the chef (unit price 28 EUR, VAT code B)

2. Partial refund:
    * refund of 1 Dry Martini (unit price 12 EUR, VAT code A)
   * sale of 1 Coca Cola (unit price 3.04 EUR, VAT code A) & 1 crème brûlée (unit price 10 EUR, VAT code
    Payment by cash.
```

Note de déclaration : le POS pili-pos ne propose pas le pattern M112A (remboursement partiel combiné à une nouvelle vente sur un même ticket). Le dossier déclare le pattern **M112B** (remboursement en événement séparé, chaîné à la vente d'origine via `fdmReference`) - voir la section M112B ci-dessous. Les fixtures M112A sont validées en *parsing* uniquement.

M112A_1_OriginalSale

Mutation invoquee : `inconnue`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	6587	OK
posDateTime	2024-10-07T13:00:00.000+02:00	OK
terminalId	TER-2-DIN	OK
deviceId	1631678d-7a85-4ac3-b296-bb4565e873fe	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-10-07	OK
ticketMedium	PAPER	OK
employeeId	84022899837	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	N	OK
fdmRef.eventCounter	81958	OK
fdmRef.totalCounter	83876	OK
shortSignature (presence)	40 caracteres hex	OK
vatCalc[] (calcul TVA par taux)	2 taux	OK

Champ	Valeur fixture	Statut

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 6587,
    "posDateTime": "2024-10-07T13:00:00.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "1631678d-7a85-4ac3-b296-bb4565e873fe",
    "terminalId": "TER-2-DIN",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-10-07",
    "ticketMedium": "PAPER",
    "employeeId": "84022899837",
    "transaction": {
      "transactionLines": [
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "10006",
            "productName": "Dry Martini",
            "departmentId": "10",
            "departmentName": "Aperitifs",
            "quantity": 2,
            "quantityType": "PIECE",
            "unitPrice": 12,
            "vats": [
              {
                "label": "A",
                "price": 24
              }
            ]
          },
          "lineTotal": 24
        },
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "22001",
            "productName": "Burger of the Chef",
            "departmentId": "22",
            "departmentName": "Main Dishes",
            "quantity": 1,
            "quantityType": "PIECE",
            "unitPrice": 28,
            "vats": [
              {
                "label": "B",
                "price": 28
              }
```

```
            ]
          },
          "lineTotal": 28
        }
      ],
      "transactionTotal": 52
    },
    "financials": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "inputMethod": "MANUAL",
        "amount": 52,
        "amountType": "PAYMENT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signSale": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 6587,
      "posDateTime": "2024-10-07T13:00:00.000+02:00",
      "terminalId": "TER-2-DIN",
      "deviceId": "1631678d-7a85-4ac3-b296-bb4565e873fe",
      "eventOperation": "SALE",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-10-07T11:00:41.000Z",
        "eventLabel": "N",
        "eventCounter": 81958,
        "totalCounter": 83876
      },
      "fdmSwVersion": "1.2.0",
      "digitalSignature": "tOKZG5v8RUeVPWu022lBIQf4gCVzAz9NlGr3JE3KyPj/e9K6fyHXRbZexqvgcSUfOlkrOS
      "shortSignature": "b6f7940dcc4694a14681471de6415defec44e02d",
      "verificationUrl": "https://gs2.be/xJYcHkRNO5UJS3hnXbo",
      "vatCalc": [
        {
          "label": "A",
          "rate": 21,
          "taxableAmount": 19.83,
          "vatAmount": 4.17,
          "totalAmount": 24,
          "outOfScope": false
        },
        {
```

```
          "label": "B",
          "rate": 12,
          "taxableAmount": 25.0,
          "vatAmount": 3.0,
          "totalAmount": 28,
          "outOfScope": false
        }
      ],
      "bufferCapacityUsed": 14.83,
      "warnings": null,
      "informations": null,
      "footer": []
    }
  }
}
```


Validation automatisee : test passant dans `crates/pos-fiscal/tests/uc_fixtures.rs` (parsing uniquement - pattern non déclaré, voir M112B).

M112A_2_PartialRefund

Mutation invoquee : `signSale`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	6603	OK
posDateTime	2024-10-07T13:48:00.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-10-07	OK
ticketMedium	PAPER	OK
employeeId	84022899837	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	N	OK
fdmRef.eventCounter	81959	OK
fdmRef.totalCounter	83877	OK
shortSignature (presence)	40 caracteres hex	OK
vatCalc[] (calcul TVA par taux)	2 taux	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 6603,
    "posDateTime": "2024-10-07T13:48:00.000+02:00",
```

```
"posSwVersion": "1.8.3",
"deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
"terminalId": "TER-1-BAR",
"bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
"bookingDate": "2024-10-07",
"ticketMedium": "PAPER",
"employeeId": "84022899837",
"transaction": {
  "transactionLines": [
    {
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "productId": "10006",
        "productName": "Dry Martini",
        "departmentId": "10",
        "departmentName": "Aperitifs",
        "quantity": -1,
        "quantityType": "PIECE",
        "negQuantityReason": "REFUND",
        "unitPrice": 12,
        "vats": [
          {
            "label": "A",
            "price": -12
          }
        ]
      },
      "lineTotal": -12
    },
    {
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "productId": "12009",
        "productName": "Coca Cola",
        "departmentId": "12",
        "departmentName": "Soft Drinks",
        "quantity": 1,
        "quantityType": "PIECE",
        "unitPrice": 3.04,
        "vats": [
          {
            "label": "A",
            "price": 3.04
          }
        ]
      },
      "lineTotal": 3.04
    },
    {
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "productId": "25008",
        "productName": "crème brûlée",
        "departmentId": "25",
        "departmentName": "Desserts",
```

```
            "quantity": 1,
            "quantityType": "PIECE",
            "unitPrice": 10,
            "vats": [
              {
                "label": "B",
                "price": 10
              }
            ]
          },
          "lineTotal": 10
        }
      ],
      "transactionTotal": 1.04
    },
    "financials": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "inputMethod": "MANUAL",
        "amount": 1.04,
        "amountType": "PAYMENT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      },
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "inputMethod": "MANUAL",
        "amount": 0.01,
        "amountType": "ROUNDING",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signSale": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 6603,
      "posDateTime": "2024-10-07T13:48:00.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "SALE",
      "fdmRef": {
```

```
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-10-07T11:48:41.000Z",
        "eventLabel": "N",
        "eventCounter": 81959,
        "totalCounter": 83877
      },
      "fdmSwVersion": "1.2.0",
      "digitalSignature": "9UrNym6R2ky3eijjqBxtnuZpRlN6FJtGtJ+W9pcNVDkiS06blUMLS7yfGvQfBWSnrL+ZZS
      "shortSignature": "e0ed61dd6d3c46bf448c1991b49a3d39ca98ee15",
      "verificationUrl": "https://gs2.be/lLApSfduWYcqKx9QQ80",
      "vatCalc": [
        {
          "label": "A",
          "rate": 21,
          "taxableAmount": -7.4,
          "vatAmount": -1.56,
          "totalAmount": -8.96,
          "outOfScope": false
        },
        {
          "label": "B",
          "rate": 12,
          "taxableAmount": 8.93,
          "vatAmount": 1.07,
          "totalAmount": 10,
          "outOfScope": false
        }
      ],
      "bufferCapacityUsed": 14.83,
      "warnings": null,
      "informations": null,
      "footer": []
    }
  }
}
```


Validation automatisee : test passant dans `crates/pos-fiscal/tests/uc_fixtures.rs` (parsing uniquement - pattern non déclaré, voir M112B).

M112B_signSale_RefundPartial

Scenario global :

```
1. Original Ticket: direct sale of
    * 2 dry Martini (unit price 12 EUR, VAT code A)
    * 1 burger of the chef (unit price 28 EUR, VAT code B)

2. Partial refund:
    * refund of 1 Dry Martini (unit price 12 EUR, VAT code A)
   * sale of 1 Coca Cola (unit price 3.04 EUR, VAT code A) & 1 crème brûlée (unit price 10 EUR, VAT code
    Payment by cash.
```

Note de déclaration : c'est CE pattern (remboursement partiel en événement séparé, chaîné à la vente d'origine via `fdmReference`) que le POS implémente et déclare. Couverture e2e applicative : `m112b_partial_refund_as_a_separate_ticket` dans `crates/pos-engine/tests/spf_e2e_fixtures.rs` (rejeu de la fixture, build `cert`) et `m112b_partial_refund_is_separate_event_chained_to_original` dans `crates/pos-engine/tests/spf_e2e_basic.rs` ; preuves générées dans `evidence/uc/M112B_*.json`

(voir « Preuve pili-pos » en fin de section).

M112B_1_OriginalSale

Mutation invoquee : `inconnue`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	6587	OK
posDateTime	2024-10-07T13:00:00.000+02:00	OK
terminalId	TER-2-DIN	OK
deviceId	1631678d-7a85-4ac3-b296-bb4565e873fe	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-10-07	OK
ticketMedium	PAPER	OK
employeeId	84022899837	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	N	OK
fdmRef.eventCounter	81958	OK
fdmRef.totalCounter	83876	OK
shortSignature (presence)	40 caracteres hex	OK
vatCalc[] (calcul TVA par taux)	2 taux	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 6587,
    "posDateTime": "2024-10-07T13:00:00.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "1631678d-7a85-4ac3-b296-bb4565e873fe",
    "terminalId": "TER-2-DIN",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-10-07",
    "ticketMedium": "PAPER",
    "employeeId": "84022899837",
    "transaction": {
      "transactionLines": [
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "10006",
            "productName": "Dry Martini",
            "departmentId": "10",
            "departmentName": "Aperitifs",
```

```
            "quantity": 2,
            "quantityType": "PIECE",
            "unitPrice": 12,
            "vats": [
              {
                "label": "A",
                "price": 24
              }
            ]
          },
          "lineTotal": 24
        },
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "22001",
            "productName": "Burger of the Chef",
            "departmentId": "22",
            "departmentName": "Main Dishes",
            "quantity": 1,
            "quantityType": "PIECE",
            "unitPrice": 28,
            "vats": [
              {
                "label": "B",
                "price": 28
              }
            ]
          },
          "lineTotal": 28
        }
      ],
      "transactionTotal": 52
    },
    "financials": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "inputMethod": "MANUAL",
        "amount": 52,
        "amountType": "PAYMENT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signSale": {
```

```
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 6587,
      "posDateTime": "2024-10-07T13:00:00.000+02:00",
      "terminalId": "TER-2-DIN",
      "deviceId": "1631678d-7a85-4ac3-b296-bb4565e873fe",
      "eventOperation": "SALE",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-10-07T11:00:41.000Z",
        "eventLabel": "N",
        "eventCounter": 81958,
        "totalCounter": 83876
      },
      "fdmSwVersion": "1.2.0",
      "digitalSignature": "tOKZG5v8RUeVPWu022lBIQf4gCVzAz9NlGr3JE3KyPj/e9K6fyHXRbZexqvgcSUfOlkrOS
      "shortSignature": "b6f7940dcc4694a14681471de6415defec44e02d",
      "verificationUrl": "https://gs2.be/xJYcHkRNO5UJS3hnXbo",
      "vatCalc": [
        {
          "label": "A",
          "rate": 21,
          "taxableAmount": 19.83,
          "vatAmount": 4.17,
          "totalAmount": 24,
          "outOfScope": false
        },
        {
          "label": "B",
          "rate": 12,
          "taxableAmount": 25.0,
          "vatAmount": 3.0,
          "totalAmount": 28,
          "outOfScope": false
        }
      ],
      "bufferCapacityUsed": 14.83,
      "warnings": [],
      "informations": null,
      "footer": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m112b_partial_refund_as_a_separate_ticket` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

M112B_2_PartialRefund

Mutation invoquee : `signSale`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	NL	OK
`posId`	CPOS0031234567	OK

Champ	Valeur fixture	Statut
posFiscalTicketNo	6603	OK
posDateTime	2024-10-07T13:48:00.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-10-07	OK
ticketMedium	PAPER	OK
employeeId	84022899837	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	N	OK
fdmRef.eventCounter	76	OK
fdmRef.totalCounter	83885	OK
shortSignature (presence)	40 caracteres hex	OK
vatCalc[] (calcul TVA par taux)	1 taux	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 6603,
    "posDateTime": "2024-10-07T13:48:00.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "terminalId": "TER-1-BAR",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-10-07",
    "ticketMedium": "PAPER",
    "employeeId": "84022899837",
    "transaction": {
      "transactionLines": [
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "10006",
            "productName": "Dry Martini",
            "departmentId": "10",
            "departmentName": "Aperitifs",
            "quantity": -1,
            "quantityType": "PIECE",
            "negQuantityReason": "REFUND",
            "unitPrice": 12,
            "vats": [
              {
                "label": "A",
                "price": -12
              }
            ]
```

```
          },
          "lineTotal": -12
        }
      ],
      "transactionTotal": -12
    },
    "financials": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "inputMethod": "MANUAL",
        "amount": -12,
        "amountType": "PAYMENT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signSale": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 6603,
      "posDateTime": "2024-10-07T13:48:00.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "SALE",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-10-07T11:48:27.000Z",
        "eventLabel": "N",
        "eventCounter": 76,
        "totalCounter": 83885
      },
      "fdmSwVersion": "1.2.0",
      "digitalSignature": "lRvExSh9UUeyZRaCZyK54Tz+WiFHb25PvXldpSsYEPOC9JN2cFEXSr9/vOP0LKCVqFekl4
      "shortSignature": "84b433a643fc935474ee744678dcfe1042fbc3c2",
      "verificationUrl": "https://gs2.be/vKRnCKepX2Yy8WiDjvU",
      "vatCalc": [
        {
          "label": "A",
          "rate": 21,
          "taxableAmount": -9.92,
          "vatAmount": -2.08,
          "totalAmount": -12,
          "outOfScope": false
        }
      ],
      "bufferCapacityUsed": 14.83,
```

```
      "warnings": null,
      "informations": null,
      "footer": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m112b_partial_refund_as_a_separate_ticket` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m112b_partia`
Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M112B_event2_N_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "153b59a3-46a3-4f50-9994-d64510f18ada",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "84022899837",
  "estNo": "2000000042",
  "fdmRef": {
    "eventCounter": 81958,
    "eventLabel": "N",
    "fdmDateTime": "2024-10-07T11:00:41.000Z",
    "fdmId": "FOD01000001",
    "totalCounter": 83876
  },
  "financials": [
    {
      "amount": -12.0,
      "amountType": "PAYMENT",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:47.817+02:00",
  "posFiscalTicketNo": 2,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "transaction": {
    "transactionLines": [
      {
        "lineTotal": -12.0,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "10",
          "departmentName": "Aperitifs",
          "negQuantityReason": "REFUND",
          "productId": "10006",
          "productName": "Dry Martini",
          "quantity": -1.0,
```

```
          "quantityType": "PIECE",
          "unitPrice": 12.0,
          "vats": [
            {
              "label": "A",
              "price": -12.0
            }
          ]
        }
      }
    ],
    "transactionTotal": -12.0
  },
  "vatNo": "BE0000000097"
}
```


Events préalables du scénario, archivés de même : `evidence/uc/M112B_event1_N_request.json`. Réponses FDM archivées : fichiers `*_response.json` correspondants.

Comparaison avec la fixture (`evidence/uc/M112B_1_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Écarts assumés :

Champ	Fixture	pili-pos	Motif
`costCenter.type`	-	TABLE	une vente en salle est encaissée depuis sa table et porte le centre de coût TABLE

Comparaison avec la fixture (`evidence/uc/M112B_2_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Écarts assumés :

Champ	Fixture	pili-pos	Motif
`fdmRef`	-	`{"eventCounter": 81958, "eventLabel": "N", "fdmDateTime": "2024-10-07T11:00:41.000Z", "fdmId": "FOD01000001", "totalCounter": 83876}`	pili-pos référence le ticket remboursé sur chaque remboursement (*FLUX_REMBOURSEMENT_FR.md*) ; le remboursement partiel de la fixture n'en porte pas

M121_signOrder

Mutation invoquee : `signOrder`

Scenario :

```
BAR: Client orders:
    2 Dry Martini (unit price 12 EUR, VAT code A)
```

```
    4 Tapas variation (unit price 3,34 EUR, VAT code B)
on table 1.
```

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	1003	OK
posDateTime	2024-07-29T13:57:03.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	PAPER	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	P	OK
fdmRef.eventCounter	22	OK
fdmRef.totalCounter	53	OK

Request (Variables.json) :


```json
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1003,
    "posDateTime": "2024-07-29T13:57:03.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "terminalId": "TER-1-BAR",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "PAPER",
    "employeeId": "75061189702",
    "costCenter": {
      "id": "T1",
      "type": "TABLE",
      "reference": "0158"
    },
    "transaction": {
      "transactionLines": [
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "10006",
            "productName": "Dry Martini",
            "departmentId": "10",
            "departmentName": "Aperitifs",
```

```
            "quantity": 2,
            "quantityType": "PIECE",
            "unitPrice": 12,
            "vats": [
              {
                "label": "A",
                "price": 24
              }
            ]
          },
          "lineTotal": 24
        },
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "28007",
            "productName": "Tapas variation",
            "departmentId": "28",
            "departmentName": "Tapas",
            "quantity": 4,
            "quantityType": "PIECE",
            "unitPrice": 3.34,
            "vats": [
              {
                "label": "B",
                "price": 13.36
              }
            ]
          },
          "lineTotal": 13.36
        }
      ],
      "transactionTotal": 37.36
    }
  }
}
```


Response FDM :


```
{
  "data": {
    "signOrder": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1003,
      "posDateTime": "2024-07-29T13:57:03.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "ORDER",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T11:57:12.000Z",
        "eventLabel": "P",
        "eventCounter": 22,
        "totalCounter": 53
      },
      "fdmSwVersion": "1.2.0",
```

```
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m121_table_order` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m121_table_o`
Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M121_event1_P_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "c4e85a25-5e9b-45fd-8b06-cc5a349dcf55",
  "costCenter": {
    "id": "T01",
    "reference": "01",
    "type": "TABLE"
  },
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:47.814+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "NONE",
  "transaction": {
    "transactionLines": [
      {
        "lineTotal": 24.0,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "10",
          "departmentName": "Aperitifs",
          "productId": "10006",
          "productName": "Dry Martini",
          "quantity": 2.0,
          "quantityType": "PIECE",
          "unitPrice": 12.0,
          "vats": [
            {
              "label": "A",
              "price": 24.0
            }
          ]
        }
      },
      {
        "lineTotal": 13.36,
```

```
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "28",
          "departmentName": "Tapas",
          "productId": "28007",
          "productName": "Tapas variation",
          "quantity": 4.0,
          "quantityType": "PIECE",
          "unitPrice": 3.34,
          "vats": [
            {
              "label": "B",
              "price": 13.36
            }
          ]
        }
      }
    ],
    "transactionTotal": 37.36
  },
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M121_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Écarts assumés :

Champ	Fixture	pili-pos	Motif
`ticketMedium`	`PAPER`	`NONE`	les commandes ne sont pas imprimées en v1.0.0 (pas de ticket cuisine) : le support est NONE

M122_signCostCenterChange

Le changement de centre de cout est disponible pour l'**attribution** : transfert d'une commande ouverte vers une table libre (*REPONSES_MANUEL_CERT_FR.md*, tableau des fonctionnalites). La division et la fusion de tables restent hors perimetre. Les valeurs ci-dessous sont celles de la fixture ; le rejeu applicatif la compare a la requete construite par pili-pos.

Mutation invoquee : `signCostCenterChange`

Scenario :

```
Order on T1 (BAR):
    1 Burger of the Chef (unit price 28 EUR, VAT code B)
    1 Dry Martini(unit price 12 EUR, VAT code A)
is moved to T5 (Dining Room)
```

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	`NL`	OK
`posId`	`CPOS0031234567`	OK

Champ	Valeur fixture	Statut
posFiscalTicketNo	1004	OK
posDateTime	2024-07-29T14:21:23.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	NONE	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	P	OK
fdmRef.eventCounter	23	OK
fdmRef.totalCounter	54	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1004,
    "posDateTime": "2024-07-29T14:21:23.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "terminalId": "TER-1-BAR",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "NONE",
    "employeeId": "75061189702",
    "transfer": {
      "from": [
        {
          "costCenter": {
            "id": "T1",
            "type": "TABLE",
            "reference": "0158"
          },
          "transaction": {
            "transactionLines": [
              {
                "lineType": "SINGLE_PRODUCT",
                "mainProduct": {
                  "productId": "22001",
                  "productName": "Burger of the Chef",
                  "departmentId": "22",
                  "departmentName": "Main Dishes",
                  "quantity": -1,
                  "negQuantityReason": "COST_CENTER_CHANGE",
                  "quantityType": "PIECE",
                  "unitPrice": 28,
```

```
            "vats": [
              {
                "label": "B",
                "price": -28
              }
            ]
          },
          "lineTotal": -28
        },
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "10006",
            "productName": "Dry Martini",
            "departmentId": "10",
            "departmentName": "Aperitifs",
            "quantity": -1,
            "negQuantityReason": "COST_CENTER_CHANGE",
            "quantityType": "PIECE",
            "unitPrice": 12,
            "vats": [
              {
                "label": "A",
                "price": -12
              }
            ]
          },
          "lineTotal": -12
        }
      ],
      "transactionTotal": -40
    }
  }
],
"to": [
  {
    "costCenter": {
      "id": "T5",
      "type": "TABLE",
      "reference": "0158"
    },
    "transaction": {
      "transactionLines": [
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "22001",
            "productName": "Burger of the Chef",
            "departmentId": "22",
            "departmentName": "Main Dishes",
            "quantity": 1,
            "quantityType": "PIECE",
            "unitPrice": 28,
            "vats": [
              {
```

```
                    "label": "B",
                    "price": 28
                  }
                ]
              },
              "lineTotal": 28
            },
            {
              "lineType": "SINGLE_PRODUCT",
              "mainProduct": {
                "productId": "10006",
                "productName": "Dry Martini",
                "departmentId": "10",
                "departmentName": "Aperitifs",
                "quantity": 1,
                "quantityType": "PIECE",
                "unitPrice": 12,
                "vats": [
                  {
                    "label": "A",
                    "price": 12
                  }
                ]
              },
              "lineTotal": 12
            }
          ],
          "transactionTotal": 40
        }
      }
    ]
  }
 }
}
```


Response FDM :


```
{
  "data": {
    "signCostCenterChange": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1004,
      "posDateTime": "2024-07-29T14:21:23.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "COST_CENTER_CHANGE",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T12:21:25.000Z",
        "eventLabel": "P",
        "eventCounter": 23,
        "totalCounter": 54
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
```

```
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m122_cost_center_change` (rejeu applicatif, build `cert`, ecart nul ; la commande suivante sur T05 garde la meme reference).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m122_cost_cer`
Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M122_event2_P_request.json`)
:


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "6b15db0f-dfbb-4a11-8da6-bb43e9c22d78",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:47.811+02:00",
  "posFiscalTicketNo": 2,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "NONE",
  "transfer": {
    "from": [
      {
        "costCenter": {
          "id": "T01",
          "reference": "01",
          "type": "TABLE"
        },
        "transaction": {
          "transactionLines": [
            {
              "lineTotal": -28.0,
              "lineType": "SINGLE_PRODUCT",
              "mainProduct": {
                "departmentId": "22",
                "departmentName": "Main Dishes",
                "negQuantityReason": "COST_CENTER_CHANGE",
                "productId": "22001",
                "productName": "Burger of the CHEF",
                "quantity": -1.0,
                "quantityType": "PIECE",
                "unitPrice": 28.0,
                "vats": [
                  {
                    "label": "B",
                    "price": -28.0
                  }
                ]
              }
            },
```

```
        {
          "lineTotal": -12.0,
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "departmentId": "10",
            "departmentName": "Aperitifs",
            "negQuantityReason": "COST_CENTER_CHANGE",
            "productId": "10006",
            "productName": "Dry Martini",
            "quantity": -1.0,
            "quantityType": "PIECE",
            "unitPrice": 12.0,
            "vats": [
              {
                "label": "A",
                "price": -12.0
              }
            ]
          }
        }
      ],
      "transactionTotal": -40.0
    }
  }
],
"to": [
  {
    "costCenter": {
      "id": "T05",
      "reference": "01",
      "type": "TABLE"
    },
    "transaction": {
      "transactionLines": [
        {
          "lineTotal": 28.0,
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "departmentId": "22",
            "departmentName": "Main Dishes",
            "productId": "22001",
            "productName": "Burger of the CHEF",
            "quantity": 1.0,
            "quantityType": "PIECE",
            "unitPrice": 28.0,
            "vats": [
              {
                "label": "B",
                "price": 28.0
              }
            ]
          }
        },
        {
          "lineTotal": 12.0,
```

```
            "lineType": "SINGLE_PRODUCT",
            "mainProduct": {
              "departmentId": "10",
              "departmentName": "Aperitifs",
              "productId": "10006",
              "productName": "Dry Martini",
              "quantity": 1.0,
              "quantityType": "PIECE",
              "unitPrice": 12.0,
              "vats": [
                {
                  "label": "A",
                  "price": 12.0
                }
              ]
            }
          }
        ],
        "transactionTotal": 40.0
      }
    }
  ]
 },
 "vatNo": "BE0000000097"
}
```


Autres events du scénario, archivés de même : `evidence/uc/M122_event1_P_request.json`, `evidence/uc/M122_event3_N_request.json`. Réponses FDM archivées : fichiers `*_response.json` correspondants.

Comparaison avec la fixture (`evidence/uc/M122_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M123_signPrebill

Mutation invoquee : `signPreBill`

Scenario :

```
Prebill for the order:
    2 Dry Martini (unit price 12 EUR, VAT code A)
    4 Tapas variation (unit price 3,34 EUR, VAT code B)
on table 1 (BAR)  is printed
```

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	`NL`	OK
`posId`	`CPOS0031234567`	OK
`posFiscalTicketNo`	`1005`	OK
`posDateTime`	`2024-07-29T14:48:03.000+02:00`	OK
`terminalId`	`TER-1-BAR`	OK
`deviceId`	`b54a614f-39cc-4a7b-bd9f-aa6b693d769c`	OK
`vatNo`	`BE0000000097`	OK
`estNo`	`2000000042`	OK

Champ	Valeur fixture	Statut
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	PAPER	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	P	OK
fdmRef.eventCounter	24	OK
fdmRef.totalCounter	55	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1005,
    "posDateTime": "2024-07-29T14:48:03.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "terminalId": "TER-1-BAR",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "PAPER",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "employeeId": "75061189702",
    "costCenter": {
      "id": "T1",
      "type": "TABLE",
      "reference": "0158"
    },
    "transaction": {
      "transactionLines": [
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "10006",
            "productName": "Dry Martini",
            "departmentId": "10",
            "departmentName": "Aperitifs",
            "quantity": 2,
            "quantityType": "PIECE",
            "unitPrice": 12,
            "vats": [
              {
                "label": "A",
                "price": 24
              }
            ]
          },
          "lineTotal": 24
        },
        {
```

```
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "28007",
            "productName": "Tapas variation",
            "departmentId": "28",
            "departmentName": "Tapas",
            "quantity": 4,
            "quantityType": "PIECE",
            "unitPrice": 3.34,
            "vats": [
              {
                "label": "B",
                "price": 13.36
              }
            ]
          },
          "lineTotal": 13.36
        }
      ],
      "transactionTotal": 37.36
    }
  }
}
```


Response FDM :


```
{
  "data": {
    "signPreBill": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1005,
      "posDateTime": "2024-07-29T14:48:03.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "PRE_BILL",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T12:48:12.000Z",
        "eventLabel": "P",
        "eventCounter": 24,
        "totalCounter": 55
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m123_pre_bill` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m123_pre_bil`
Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M123_event2_P_request.json`)

:


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "b49ede41-611e-4057-9971-1b8d3e8c4be7",
  "costCenter": {
    "id": "T01",
    "reference": "O1",
    "type": "TABLE"
  },
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:47.820+02:00",
  "posFiscalTicketNo": 2,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "transaction": {
    "transactionLines": [
      {
        "lineTotal": 24.0,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "10",
          "departmentName": "Aperitifs",
          "productId": "10006",
          "productName": "Dry Martini",
          "quantity": 2.0,
          "quantityType": "PIECE",
          "unitPrice": 12.0,
          "vats": [
            {
              "label": "A",
              "price": 24.0
            }
          ]
        }
      },
      {
        "lineTotal": 13.36,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "28",
          "departmentName": "Tapas",
          "productId": "28007",
          "productName": "Tapas variation",
          "quantity": 4.0,
          "quantityType": "PIECE",
          "unitPrice": 3.34,
          "vats": [
            {
              "label": "B",
              "price": 13.36
```

```
            }
          ]
        }
      }
    ],
    "transactionTotal": 37.36
  },
  "vatNo": "BE0000000097"
}
```


Events préalables du scénario, archivés de même : `evidence/uc/M123_event1_P_request.json`. Réponses FDM archivées : fichiers `*_response.json` correspondants.

Comparaison avec la fixture (`evidence/uc/M123_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M130_signMoneyInOut

Mutation invoquee : `signMoneyInOut`

Scenario :

`100 EUR cash is placed in drawer1`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	`NL`	OK
`posId`	`CPOS0031234567`	OK
`posFiscalTicketNo`	`1006`	OK
`posDateTime`	`2024-07-29T15:07:09.000+02:00`	OK
`terminalId`	`TER-1-BAR`	OK
`deviceId`	`b54a614f-39cc-4a7b-bd9f-aa6b693d769c`	OK
`vatNo`	`BE0000000097`	OK
`estNo`	`2000000042`	OK
`posSwVersion`	`1.8.3`	OK
`bookingPeriodId`	`dffcd829-a0e5-41ca-a0ae-9eb887f95637`	OK
`bookingDate`	`2024-07-29`	OK
`ticketMedium`	`NONE`	OK
`employeeId`	`75061189702`	OK
`fdmRef.fdmId`	`FOD01000001`	OK
`fdmRef.eventLabel`	`F`	OK
`fdmRef.eventCounter`	`6`	OK
`fdmRef.totalCounter`	`56`	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1006,
```

```
    "posDateTime": "2024-07-29T15:07:09.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "NONE",
    "employeeId": "75061189702",
    "financials": [
      {
        "id": "1",
        "name": "CASH",
        "type": "CASH",
        "amount": 100,
        "inputMethod": "MANUAL",
        "amountType": "MONEY_IN_OUT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signMoneyInOut": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1006,
      "posDateTime": "2024-07-29T15:07:09.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "MONEY_IN_OUT",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T13:07:12.000Z",
        "eventLabel": "F",
        "eventCounter": 6,
        "totalCounter": 56
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m130_money_in` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m130_money_i`

Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M130_event1_F_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "15873b12-4e57-4f82-9228-b56598dd9656",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "financials": [
    {
      "amount": 100.0,
      "amountType": "MONEY_IN_OUT",
      "id": "MOV1",
      "inputMethod": "MANUAL",
      "name": "Fond de caisse",
      "type": "CASH"
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:47.804+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "NONE",
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M130_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M131_signDrawerOpen

Mutation invoquee : `signDrawerOpen`

Scenario :

`Drawer 1 is opened without any action`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	1007	OK
posDateTime	2024-07-29T15:12:04.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	NONE	OK

Champ	Valeur fixture	Statut
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	F	OK
fdmRef.eventCounter	7	OK
fdmRef.totalCounter	57	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1007,
    "posDateTime": "2024-07-29T15:12:04.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "NONE",
    "employeeId": "75061189702",
    "drawer": {
      "id": "1",
      "name": "Drawer 1"
    }
  }
}
```


Response FDM :


```
{
  "data": {
    "signDrawerOpen": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1007,
      "posDateTime": "2024-07-29T15:12:04.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "DRAWER_OPEN",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T13:12:05.000Z",
        "eventLabel": "F",
        "eventCounter": 7,
        "totalCounter": 57
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
```

```
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m131_drawer_open` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m131_drawer_` Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M131_event1_F_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "6c96a484-cdac-41b9-b20a-a09cfe05f0fc",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "drawer": {
    "id": "DRAWER1",
    "name": "Tiroir principal"
  },
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:48.479+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "NONE",
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M131_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M132_signPaymentCorrection

La correction de paiement est disponible pour une ligne de paiement saisie a la main d'une vente de la journee en cours (*REPONSES_MANUEL_CERT_FR.md*, tableau des fonctionnalites). Les valeurs ci-dessous sont celles de la fixture ; le rejeu applicatif la compare a la requete construite par pili-pos. Un ecart declare : la fixture corrige vers une carte de credit (`CARD_CREDIT`), alors que, sans terminal relie, une carte saisie a la main est toujours `CARD_DEBIT` (*REPONSES_MANUEL_CERT_FR.md* §III.F). Deux differences ne sont pas comparees (la comparaison porte sur le type, le montant et le type de montant de chaque ligne) : la nouvelle ligne de carte part en saisie manuelle (`inputMethod` MANUAL, la fixture a AUTOMATIC) et la ligne d'especes n'indique pas de tiroir (champ optionnel, la fixture nomme le tiroir 1).

Mutation invoquee : `signPaymentCorrection`

Scenario :

```
PaymentCorrection

Payment of N14/50 (total to be paid 52 EUR) was made by CARD_CREDIT (Mastercard via EFT) and not in cas
```

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	1008	OK
posDateTime	2024-07-29T15:22:06.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	NONE	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	F	OK
fdmRef.eventCounter	8	OK
fdmRef.totalCounter	58	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1008,
    "posDateTime": "2024-07-29T15:22:06.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "NONE",
    "employeeId": "75061189702",
    "fdmRef": {
      "fdmId": "FOD01000001",
      "fdmDateTime": "2024-07-29T11:35:09.000Z",
      "eventLabel": "N",
      "eventCounter": 14,
      "totalCounter": 50
    },
    "financials": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "inputMethod": "MANUAL",
        "amount": -52,
        "amountType": "PAYMENT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
```

```
      },
      {
        "id": "2",
        "name": "MASTERCARD",
        "type": "CARD_CREDIT",
        "inputMethod": "AUTOMATIC",
        "amount": 52,
        "amountType": "PAYMENT"
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signPaymentCorrection": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1008,
      "posDateTime": "2024-07-29T15:22:06.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "PAYMENT_CORRECTION",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T13:22:06.000Z",
        "eventLabel": "F",
        "eventCounter": 8,
        "totalCounter": 58
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Ecarts declares du rejeu M132. La comparaison automatique ne porte que sur le type, le montant et le type de montant (`amountType`) de chaque ligne de paiement. Un ecart est declare : la nouvelle ligne de carte est `CARD_DEBIT` (la fixture a `CARD_CREDIT`), car sans terminal de paiement relie l'operateur ne peut pas distinguer debit et credit. Deux differences ne sont pas comparees : la nouvelle ligne de carte part en saisie manuelle (`inputMethod` `MANUAL`, la fixture a `AUTOMATIC`, puisqu'elle suppose un terminal) et la ligne d'especes n'indique pas de tiroir (champ optionnel, la fixture nomme le tiroir 1).

Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m132_payment_correction` (rejeu applicatif, build `cert`, un seul ecart : le type de la carte, `CARD_DEBIT` au lieu de `CARD_CREDIT`).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m132_payment_` Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M132_event2_F_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "94db736e-23cd-4cd2-a28f-e1e085bb6d54",
```

```
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "fdmRef": {
    "eventCounter": 14,
    "eventLabel": "N",
    "fdmDateTime": "2024-07-29T11:35:09.000Z",
    "fdmId": "FOD01000001",
    "totalCounter": 50
  },
  "financials": [
    {
      "amount": -52.0,
      "amountType": "PAYMENT",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    },
    {
      "amount": 52.0,
      "amountType": "PAYMENT",
      "id": "2",
      "inputMethod": "MANUAL",
      "name": "Card",
      "type": "CARD_DEBIT"
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:53.001+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "NONE",
  "vatNo": "BE0000000097"
}
```


Events préalables du scénario, archivés de même : `evidence/uc/M132_event1_N_request.json`. Réponses FDM archivées : fichiers `*_response.json` correspondants.

Comparaison avec la fixture (`evidence/uc/M132_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Écarts assumés :

Champ	Fixture	pili-pos	Motif
`financials[0].type`	`CARD_CREDIT`	`CARD_DEBIT`	sans terminal relié, l'opérateur ne distingue pas débit et crédit : une carte saisie à la main est CARD_DEBIT, MANUAL (*RE-PONSES_MANUEL_CERT_FR.md* §III.F) ; la comparaison ne porte que sur le type, le montant et le type de montant, la saisie MANUAL (AUTOMATIC dans la fixture) n'est donc pas comparée

M133_signMoneyInOut_DrawerDeclaration

Mutation invoquee : `signMoneyInOut`

Scenario :

`Declaration of 1000 EUR cash in drawer 1`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	NL	OK
`posId`	CPOS0031234567	OK
`posFiscalTicketNo`	1009	OK
`posDateTime`	2024-07-29T15:39:39.000+02:00	OK
`terminalId`	TER-1-BAR	OK
`deviceId`	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
`vatNo`	BE0000000097	OK
`estNo`	2000000042	OK
`posSwVersion`	1.8.3	OK
`bookingPeriodId`	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
`bookingDate`	2024-07-29	OK
`ticketMedium`	NONE	OK
`employeeId`	75061189702	OK
`fdmRef.fdmId`	FOD01000001	OK
`fdmRef.eventLabel`	F	OK
`fdmRef.eventCounter`	9	OK
`fdmRef.totalCounter`	59	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1009,
    "posDateTime": "2024-07-29T15:39:39.000+02:00",
```

```
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "NONE",
    "employeeId": "75061189702",
    "financials": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "amount": 1000,
        "inputMethod": "MANUAL",
        "amountType": "DRAWER_DECLARATION",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signMoneyInOut": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1009,
      "posDateTime": "2024-07-29T15:39:39.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "MONEY_IN_OUT",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T13:39:45.000Z",
        "eventLabel": "F",
        "eventCounter": 9,
        "totalCounter": 59
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m133_drawer_declaration` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m133_drawer_`
Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M133_event1_F_request.json`)

:


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "d8f6da0b-79cd-464a-8340-e8f7317ce19f",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "financials": [
    {
      "amount": 1000.0,
      "amountType": "DRAWER_DECLARATION",
      "id": "MOV1",
      "inputMethod": "MANUAL",
      "name": "Declaration tiroir",
      "type": "CASH"
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:48.472+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "NONE",
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M133_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M140_signWorkIn

Mutation invoquee : `signWorkIn`

Scenario :

`Registration of "Work In" of employee (employeeId 80113078968)`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234568	OK
posFiscalTicketNo	1010	OK
posDateTime	2024-07-29T15:50:39.000+02:00	OK
terminalId	POS-2-REC	OK
deviceId	80:1A:A6:75:66:FC	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	NONE	OK
employeeId	80113078968	OK

Champ	Valeur fixture	Statut
`fdmRef.fdmId`	`FOD01000002`	OK
`fdmRef.eventLabel`	`S`	OK
`fdmRef.eventCounter`	`7`	OK
`fdmRef.totalCounter`	`30`	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234568",
    "posFiscalTicketNo": 1010,
    "posDateTime": "2024-07-29T15:50:39.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "POS-2-REC",
    "deviceId": "80:1A:A6:75:66:FC",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "NONE",
    "employeeId": "80113078968"
  }
}
```


Response FDM :


```
{
  "data": {
    "signWorkIn": {
      "posId": "CPOS0031234568",
      "posFiscalTicketNo": 1010,
      "posDateTime": "2024-07-29T15:50:39.000+02:00",
      "terminalId": "POS-2-REC",
      "deviceId": "80:1A:A6:75:66:FC",
      "eventOperation": "WORK_IN",
      "fdmRef": {
        "fdmId": "FOD01000002",
        "fdmDateTime": "2024-07-29T13:50:39.000Z",
        "eventLabel": "S",
        "eventCounter": 7,
        "totalCounter": 30
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m140_work_in` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m140_work_in`. Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M140_event1_S_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "9c026fff-3842-405d-8fec-adaab07145eb",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "80113078968",
  "estNo": "2000000042",
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:48.483+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "NONE",
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M140_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M141_signWorkOut

Mutation invoquee : `signWorkOut`

Scenario :

`Registration of "Work Out" of employee (employeeId 84022899837)`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	NL	OK
`posId`	CPOS0031234567	OK
`posFiscalTicketNo`	1011	OK
`posDateTime`	2024-07-29T15:59:39.000+02:00	OK
`terminalId`	TER-1-BAR	OK
`deviceId`	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
`vatNo`	BE0000000097	OK
`estNo`	2000000042	OK
`posSwVersion`	1.8.3	OK
`bookingPeriodId`	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
`bookingDate`	2024-07-29	OK
`ticketMedium`	NONE	OK
`employeeId`	84022899837	OK
`fdmRef.fdmId`	FOD01000001	OK
`fdmRef.eventLabel`	S	OK
`fdmRef.eventCounter`	8	OK
`fdmRef.totalCounter`	61	OK

Request (Variables.json) :

```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1011,
    "posDateTime": "2024-07-29T15:59:39.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "NONE",
    "employeeId": "84022899837"
  }
}
```


Response FDM :


```
{
  "data": {
    "signWorkOut": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1011,
      "posDateTime": "2024-07-29T15:59:39.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "WORK_OUT",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T13:59:43.000Z",
        "eventLabel": "S",
        "eventCounter": 8,
        "totalCounter": 61
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m141_work_out` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m141_work_ou`
Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M141_event1_S_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "47656434-bd49-4a16-9df7-e74b32aa9eb5",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "84022899837",
```

```
  "estNo": "2000000042",
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:48.511+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "NONE",
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M141_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M150_signInvoice

Mutation invoquee : `signInvoice`

Scenario :

```
Invoice for customer (Id 100 - customerVatNo BE0308357159)
for tickets
    N 14/50
    N 16/52
```

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	1012	OK
posDateTime	2024-07-29T16:22:39.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	PAPER	OK
employeeId	75061189702	OK
customerVatNo	BE0308357159	OK
costCenter.id	100	OK
costCenter.type	CUSTOMER	OK
costCenter.reference	FIN	OK
invoiceNo	20240005	OK
fdmRefs[0].fdmId	FOD01000001	OK
fdmRefs[0].eventLabel	N	OK
fdmRefs[0].eventCounter	14	OK
fdmRefs[0].totalCounter	50	OK
fdmRefs[1].fdmId	FOD01000001	OK
fdmRefs[1].eventLabel	N	OK
fdmRefs[1].eventCounter	16	OK
fdmRefs[1].totalCounter	52	OK

Champ	Valeur fixture	Statut
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	I	OK
fdmRef.eventCounter	1	OK
fdmRef.totalCounter	62	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1012,
    "posDateTime": "2024-07-29T16:22:39.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "PAPER",
    "employeeId": "75061189702",
    "customerVatNo": "BE0308357159",
    "costCenter": {
      "id": "100",
      "type": "CUSTOMER",
      "reference": "FIN"
    },
    "invoiceNo": "20240005",
    "fdmRefs": [
      {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-04-29T11:35:09.000Z",
        "eventLabel": "N",
        "eventCounter": 14,
        "totalCounter": 50
      },
      {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T11:48:04.000Z",
        "eventLabel": "N",
        "eventCounter": 16,
        "totalCounter": 52
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signInvoice": {
      "posId": "CPOS0031234567",
```

```
      "posFiscalTicketNo": 1012,
      "posDateTime": "2024-07-29T16:22:39.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "INVOICE",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T14:22:41.000Z",
        "eventLabel": "I",
        "eventCounter": 1,
        "totalCounter": 62
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Rejeu applicatif : `crates/pos-engine/tests/spf_e2e_fixtures.rs::m150_invoice` (build `cert`, comme tous les UC ci-dessus). Le rejeu fait deux ventes (événements N), établit la facture par la commande de l'écran de facturation et compare la requête archivée à la fixture. Champs métier égaux (`costCenter.type` et `costCenter.reference`, `customerVatNo`, libellés des références). Chaque `fdmRefs[i]` (`fdmId`, `fdmDateTime`, `eventLabel`, `eventCounter`, `totalCounter`) est comparé à la référence que le module de contrôle a répondue pour la vente correspondante, dans l'ordre. Différences voulues, propres à la session de test : identifiant du centre de coût (`C` suivi du numéro du client, contre `100` dans le scénario), numéro de facture (année et séquence continue, `20260001` à la place de `20240005`) et compteurs des références. Voir *FACTURATION_PEPPOL_FR.md*.

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m150_invoice`. Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M150_event3_I_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "8817ade4-f83b-428e-aea5-213b6f31208d",
  "costCenter": {
    "id": "C1",
    "reference": "FIN",
    "type": "CUSTOMER"
  },
  "customerVatNo": "BE0308357159",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "fdmRefs": [
    {
      "eventCounter": 15,
      "eventLabel": "N",
      "fdmDateTime": "2024-07-29T11:00:08.000Z",
      "fdmId": "FOD01000001",
      "totalCounter": 59
    },
    {
      "eventCounter": 15,
```

```
      "eventLabel": "N",
      "fdmDateTime": "2024-07-29T11:00:08.000Z",
      "fdmId": "FOD01000001",
      "totalCounter": 59
    }
  ],
  "invoiceNo": "20260001",
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:48.624+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "vatNo": "BE0000000097"
}
```


Events préalables du scénario, archivés de même : `evidence/uc/M150_event1_N_request.json`, `evidence/uc/M150_event2_N_request.json`. Réponses FDM archivées : fichiers `*_response.json` correspondants.

Comparaison avec la fixture (`evidence/uc/M150_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M160_signCopy_Sale

Mutation invoquee : `signCopy`

Scenario :

`Print of a copy of ticket (sale) N 16/52`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	NL	OK
`posId`	CPOS0031234567	OK
`posFiscalTicketNo`	1013	OK
`posDateTime`	2024-07-29T16:25:39.000+02:00	OK
`terminalId`	TER-1-BAR	OK
`deviceId`	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
`vatNo`	BE0000000097	OK
`estNo`	2000000042	OK
`posSwVersion`	1.8.3	OK
`bookingPeriodId`	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
`bookingDate`	2024-07-29	OK
`ticketMedium`	PAPER	OK
`employeeId`	75061189702	OK
`fdmRef.fdmId`	FOD01000001	OK
`fdmRef.eventLabel`	C	OK
`fdmRef.eventCounter`	2	OK
`fdmRef.totalCounter`	63	OK

Request (Variables.json) :

```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1013,
    "posDateTime": "2024-07-29T16:25:39.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "PAPER",
    "employeeId": "75061189702",
    "fdmRef": {
      "fdmId": "FOD01000001",
      "fdmDateTime": "2024-07-29T11:48:04.000Z",
      "eventLabel": "N",
      "eventCounter": 16,
      "totalCounter": 52
    }
  }
}
```


Response FDM :


```
{
  "data": {
    "signCopy": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1013,
      "posDateTime": "2024-07-29T16:25:39.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "COPY",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T14:25:45.000Z",
        "eventLabel": "C",
        "eventCounter": 2,
        "totalCounter": 63
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m160_copy_of_a_sale` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m160_copy_of_`

Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M160_event2_C_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "76058c32-1a33-4041-a8d7-e601eafc46ab",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "fdmRef": {
    "eventCounter": 16,
    "eventLabel": "N",
    "fdmDateTime": "2024-07-29T11:48:04.000Z",
    "fdmId": "FOD01000001",
    "totalCounter": 52
  },
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:48.738+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "vatNo": "BE0000000097"
}
```


Events préalables du scénario, archivés de même : `evidence/uc/M160_event1_N_request.json`. Réponses FDM archivées : fichiers `*_response.json` correspondants.

Comparaison avec la fixture (`evidence/uc/M160_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M161_signCopy_Order

Mutation invoquee : `signCopy`

Scenario :

`Print of a copy of order P 22/53`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	1014	OK
posDateTime	2024-07-29T16:30:39.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	PAPER	OK

Champ	Valeur fixture	Statut
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	C	OK
fdmRef.eventCounter	3	OK
fdmRef.totalCounter	64	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1014,
    "posDateTime": "2024-07-29T16:30:39.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "PAPER",
    "employeeId": "75061189702",
    "fdmRef": {
      "fdmId": "FOD01000001",
      "fdmDateTime": "2024-07-29T12:57:12.000Z",
      "eventLabel": "P",
      "eventCounter": 22,
      "totalCounter": 53
    }
  }
}
```


Response FDM :


```
{
  "data": {
    "signCopy": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1014,
      "posDateTime": "2024-07-29T16:30:39.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "COPY",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T14:30:42.000Z",
        "eventLabel": "C",
        "eventCounter": 3,
        "totalCounter": 64
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
```

```
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m161_copy_of_an_order` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m161_copy_of_` Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M161_event2_C_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "6c96d35b-1794-45d1-b685-b0792c7a57a9",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "fdmRef": {
    "eventCounter": 22,
    "eventLabel": "P",
    "fdmDateTime": "2024-07-29T12:57:12.000Z",
    "fdmId": "FOD01000001",
    "totalCounter": 53
  },
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:48.443+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "vatNo": "BE0000000097"
}
```


Events préalables du scénario, archivés de même : `evidence/uc/M161_event1_P_request.json`. Réponses FDM archivées : fichiers `*_response.json` correspondants.

Comparaison avec la fixture (`evidence/uc/M161_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M170_signWorkIn_Training

Mutation invoquee : `signWorkIn`

Scenario :

`Registration of "Work In" of an employee (employeeId 75061189702) in training modus`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	NL	OK
`posId`	`CPOS0031234567`	OK

Champ	Valeur fixture	Statut
posFiscalTicketNo	1014	OK
posDateTime	2024-07-29T17:00:39.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	NONE	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	T	OK
fdmRef.eventCounter	3	OK
fdmRef.totalCounter	65	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1014,
    "posDateTime": "2024-07-29T17:00:39.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "NONE",
    "employeeId": "75061189702"
  },
  "training": true
}
```


Response FDM :


```
{
  "data": {
    "signWorkIn": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1014,
      "posDateTime": "2024-07-29T17:00:39.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "WORK_IN",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T15:00:41.000Z",
        "eventLabel": "T",
        "eventCounter": 3,
        "totalCounter": 65
```

```
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m170_work_in_in_training` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m170_work_in_` Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M170_event1_T_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "8a0e4ed2-e9e2-487c-8bc7-e2c47d23c81a",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:48.958+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "NONE",
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M170_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Aucun écart.

M171B_signSale_Training

Mutation invoquee : `signSale`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	NL	OK
`posId`	CPOS0031234567	OK
`posFiscalTicketNo`	1015	OK
`posDateTime`	2024-07-29T17:10:39.000+02:00	OK
`terminalId`	TER-1-BAR	OK
`deviceId`	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
`vatNo`	BE0000000097	OK
`estNo`	2000000042	OK
`posSwVersion`	1.8.3	OK
`bookingPeriodId`	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
`bookingDate`	2024-07-29	OK

Champ	Valeur fixture	Statut
ticketMedium	PAPER	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	T	OK
fdmRef.eventCounter	4	OK
fdmRef.totalCounter	64	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1015,
    "posDateTime": "2024-07-29T17:10:39.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "PAPER",
    "employeeId": "75061189702",
    "transaction": {
      "transactionLines": [
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "12007",
            "productName": "Perrier",
            "departmentId": "12",
            "departmentName": "Soft Drinks",
            "quantity": 2,
            "quantityType": "PIECE",
            "unitPrice": 2.5,
            "vats": [
              {
                "label": "A",
                "price": 5
              }
            ]
          },
          "lineTotal": 5
        },
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "22001",
            "productName": "Burger of the Chef",
            "departmentId": "22",
            "departmentName": "Main Dishes",
            "quantity": 1,
            "quantityType": "PIECE",
```

```
            "unitPrice": 28,
            "vats": [
              {
                "label": "B",
                "price": 28
              }
            ]
          },
          "lineTotal": 28
        }
      ],
      "transactionTotal": 33
    },
    "financials": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "inputMethod": "MANUAL",
        "amount": 33,
        "amountType": "PAYMENT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  },
  "training": true
}
```


Response FDM :


```
{
  "data": {
    "signSale": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1015,
      "posDateTime": "2024-07-29T17:10:39.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "fdmSwVersion": "1.2.0",
      "eventOperation": "SALE",
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "bufferCapacityUsed": 0.25,
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T15:10:41.000Z",
        "eventLabel": "T",
        "eventCounter": 4,
        "totalCounter": 64
      },
      "warnings": null,
      "informations": null
    }
  }
```

```
}
```

Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m171b_sale_in_training_with_skipped_fields` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m171b_sale_i`
Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M171B_event1_T_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "91a4ee31-d9ac-410f-b179-65c4c8be4517",
  "costCenter": {
    "id": "T01",
    "reference": "01",
    "type": "TABLE"
  },
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "financials": [
    {
      "amount": 33.0,
      "amountType": "PAYMENT",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:48.814+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "transaction": {
    "transactionLines": [
      {
        "lineTotal": 5.0,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "12",
          "departmentName": "Soft Drinks",
          "productId": "12007",
          "productName": "Perrier",
          "quantity": 2.0,
          "quantityType": "PIECE",
          "unitPrice": 2.5,
          "vats": [
            {
              "label": "A",
              "price": 5.0
            }
          ]
```

```
        }
      },
      {
        "lineTotal": 28.0,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "22",
          "departmentName": "Main Dishes",
          "productId": "22001",
          "productName": "Burger of the CHEF",
          "quantity": 1.0,
          "quantityType": "PIECE",
          "unitPrice": 28.0,
          "vats": [
            {
              "label": "B",
              "price": 28.0
            }
          ]
        }
      }
    ],
    "transactionTotal": 33.0
  },
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M171B_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Écarts assumés :

Champ	Fixture	pili-pos	Motif
`costCenter.type`	-	`TABLE`	une vente en salle est encaissée depuis sa table et porte le centre de coût TABLE

M171_signSale_Training

Mutation invoquee : `signSale`

Scenario :

```
Sale of
    2 Perrier (unit price 2.50 EUR, VAT code A)
    1 Burger of the Chef (unit price 28.00 EUR, VAT code B)
in training mode. Paymentmethod: CASH
```

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	`NL`	OK
`posId`	`CPOS0031234567`	OK
`posFiscalTicketNo`	`1015`	OK

Champ	Valeur fixture	Statut
posDateTime	2024-07-29T17:10:39.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	PAPER	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	T	OK
fdmRef.eventCounter	4	OK
fdmRef.totalCounter	64	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1015,
    "posDateTime": "2024-07-29T17:10:39.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "PAPER",
    "employeeId": "75061189702",
    "transaction": {
      "transactionLines": [
        {
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "12007",
            "productName": "Perrier",
            "departmentId": "12",
            "departmentName": "Soft Drinks",
            "quantity": 2,
            "quantityType": "PIECE",
            "unitPrice": 2.5,
            "vats": [
              {
                "label": "A",
                "price": 5
              }
            ]
          },
          "lineTotal": 5
        },
        {
```

```
          "lineType": "SINGLE_PRODUCT",
          "mainProduct": {
            "productId": "22001",
            "productName": "Burger of the Chef",
            "departmentId": "22",
            "departmentName": "Main Dishes",
            "quantity": 1,
            "quantityType": "PIECE",
            "unitPrice": 28,
            "vats": [
              {
                "label": "B",
                "price": 28
              }
            ]
          },
          "lineTotal": 28
        }
      ],
      "transactionTotal": 33
    },
    "financials": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "inputMethod": "MANUAL",
        "amount": 33,
        "amountType": "PAYMENT",
        "drawer": {
          "id": "1",
          "name": "Drawer 1"
        }
      }
    ]
  },
  "training": true
}
```


Response FDM :


```
{
  "data": {
    "signSale": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1015,
      "posDateTime": "2024-07-29T17:10:39.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "fdmSwVersion": "1.2.0",
      "eventOperation": "SALE",
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "shortSignature": null,
      "verificationUrl": null,
      "bufferCapacityUsed": 0.25,
      "fdmRef": {
```

```
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T15:10:41.000Z",
        "eventLabel": "T",
        "eventCounter": 4,
        "totalCounter": 64
      },
      "vatCalc": null,
      "warnings": null,
      "informations": null
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m171_sale_in_training` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m171_sale_in_` Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M171_event1_T_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "5f4921cb-553f-4fe3-b0df-542132be2375",
  "costCenter": {
    "id": "T01",
    "reference": "01",
    "type": "TABLE"
  },
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "financials": [
    {
      "amount": 33.0,
      "amountType": "PAYMENT",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:48.961+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "transaction": {
    "transactionLines": [
      {
        "lineTotal": 5.0,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "12",
          "departmentName": "Soft Drinks",
```

```
          "productId": "12007",
          "productName": "Perrier",
          "quantity": 2.0,
          "quantityType": "PIECE",
          "unitPrice": 2.5,
          "vats": [
            {
              "label": "A",
              "price": 5.0
            }
          ]
        }
      },
      {
        "lineTotal": 28.0,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "22",
          "departmentName": "Main Dishes",
          "productId": "22001",
          "productName": "Burger of the CHEF",
          "quantity": 1.0,
          "quantityType": "PIECE",
          "unitPrice": 28.0,
          "vats": [
            {
              "label": "B",
              "price": 28.0
            }
          ]
        }
      }
    ],
    "transactionTotal": 33.0
  },
  "vatNo": "BE0000000097"
}
```


Comparaison avec la fixture (`evidence/uc/M171_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Écarts assumés :

Champ	Fixture	pili-pos	Motif
`costCenter.type`	-	TABLE	une vente en salle est encaissée depuis sa table et porte le centre de coût TABLE

M172_signCopy_Training

Mutation invoquee : `signCopy`

Scenario :

```
In training mode: print of a copy of ticket (sale) N 35/73
```

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
language	NL	OK
posId	CPOS0031234567	OK
posFiscalTicketNo	1016	OK
posDateTime	2024-07-29T17:10:39.000+02:00	OK
terminalId	TER-1-BAR	OK
deviceId	b54a614f-39cc-4a7b-bd9f-aa6b693d769c	OK
vatNo	BE0000000097	OK
estNo	2000000042	OK
posSwVersion	1.8.3	OK
bookingPeriodId	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
bookingDate	2024-07-29	OK
ticketMedium	NONE	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	T	OK
fdmRef.eventCounter	7	OK
fdmRef.totalCounter	67	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 1016,
    "posDateTime": "2024-07-29T17:10:39.000+02:00",
    "posSwVersion": "1.8.3",
    "terminalId": "TER-1-BAR",
    "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-29",
    "ticketMedium": "NONE",
    "employeeId": "75061189702",
    "fdmRef": {
      "fdmId": "FOD01000002",
      "fdmDateTime": "2024-07-03T16:40:39.000Z",
      "eventLabel": "N",
      "eventCounter": 35,
      "totalCounter": 73
    }
  },
  "training": true
}
```


Response FDM :


```
{
  "data": {
    "signCopy": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 1016,
```

```
      "posDateTime": "2024-07-29T17:10:39.000+02:00",
      "terminalId": "TER-1-BAR",
      "deviceId": "b54a614f-39cc-4a7b-bd9f-aa6b693d769c",
      "eventOperation": "COPY",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-29T15:10:45.000Z",
        "eventLabel": "T",
        "eventCounter": 7,
        "totalCounter": 67
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_fixtures.rs::m172_copy_in_training` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - rejeu de la fixture par `crates/pos-engine/tests/spf_e2e_fixtures.rs::m172_copy_in_` Requête **construite et archivée par pili-pos** pour l'event de l'UC (`evidence/uc/M172_event2_T_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "1946ccf4-d769-499a-8616-8475894d9816",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "fdmRef": {
    "eventCounter": 35,
    "eventLabel": "N",
    "fdmDateTime": "2024-07-03T16:40:39.000Z",
    "fdmId": "FOD01000002",
    "totalCounter": 73
  },
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:48.972+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "vatNo": "BE0000000097"
}
```


Events préalables du scénario, archivés de même : `evidence/uc/M172_event1_N_request.json`. Réponses FDM archivées : fichiers `*_response.json` correspondants.

Comparaison avec la fixture (`evidence/uc/M172_check.json`) : champs métier identiques à la fixture ; champs d'identification (posId, deviceId, numéro de ticket, dates, période de booking) contrôlés sur leur format ; libellés présents.

Écarts assumés :

Champ	Fixture	pili-pos	Motif
`ticketMedium`	`NONE`	`PAPER`	les copies sont imprimées aussi en mode formation : le support est PAPER

Q100_Status

Mutation invoquee : `inconnue`

Aucun champ a auto-cocher : voir fixtures.

Request (Variables.json) : aucune fixture trouvee

Response FDM :


```
{
  "data": {
    "status": {
      "__typename": "FdmStatus",
      "initialized": true,
      "device": {
        "bufferCapacityUsed": 10.02,
        "fdmDateTime": "2025-08-10T10:00:39.000Z",
        "fdmId": "FOD01000001",
        "fdmSwVersion": "1.2.0"
      },
      "errors": [],
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-fiscal/tests/client_status.rs` (requete de statut du client FDM).

Q110_Status_multi_lang_messages

Mutation invoquee : `inconnue`

Aucun champ a auto-cocher : voir fixtures.

Request (Variables.json) : aucune fixture trouvee

Response FDM :


```
{
  "data": {
    "status": {
      "__typename": "FdmStatus",
      "initialized": true,
      "device": {
        "__typename": "Device",
        "bufferCapacityUsed": 10.02,
        "fdmDateTime": "2025-08-10T10:00:39.000Z",
        "fdmId": "FOD01000001",
```

```
        "fdmSwVersion": "1.2.0"
      },
      "errors": [],
      "warnings": [],
      "informations": []
    },
    "messagesFR": {
      "errors": [],
      "warnings": [],
      "informations": []
    },
    "messagesNL": {
      "errors": [],
      "warnings": [],
      "informations": []
    },
    "messagesDE": {
      "errors": [],
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/uc_fixtures.rs` (lecture de la fixture) et `crates/pos-fiscal/tests/client_status.rs` (requete de statut du client FDM).

Reports v1.3 (5 UC)

M180_signReportTurnoverX

Mutation invoquee : `signReportTurnoverX`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	`NL`	OK
`posId`	`CPOS0031234567`	OK
`posFiscalTicketNo`	`208`	OK
`posDateTime`	`2024-07-30T13:00:35.000+02:00`	OK
`terminalId`	`POS-1-BAR`	OK
`deviceId`	`F0:1D:3D:CC:13:C8`	OK
`vatNo`	`BE0000000097`	OK
`estNo`	`2000000042`	OK
`posSwVersion`	`1.8.3`	OK
`bookingPeriodId`	`dffcd829-a0e5-41ca-a0ae-9eb887f95637`	OK
`bookingDate`	`2024-07-30`	OK
`ticketMedium`	`PAPER`	OK
`employeeId`	`75061189702`	OK
`fdmRef.fdmId`	`FOD01000001`	OK
`fdmRef.eventLabel`	`R`	OK
`fdmRef.eventCounter`	`3`	OK
`fdmRef.totalCounter`	`57`	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 208,
    "posDateTime": "2024-07-30T13:00:35.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "F0:1D:3D:CC:13:C8",
    "terminalId": "POS-1-BAR",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-30",
    "ticketMedium": "PAPER",
    "employeeId": "75061189702",
    "posDevices": [
      {
        "posId": "CPOS0031234567",
        "terminalId": "TER-1-BAR",
        "firstPosDateTime": "2024-07-30T08:30:12.000+02:00",
        "lastPosDateTime": "2024-07-30T08:40:12.000+02:00"
      },
      {
        "posId": "CPOS0031234567",
        "terminalId": "TER-2-DIN",
        "firstPosDateTime": "2024-07-30T11:45:12.000+02:00",
        "lastPosDateTime": "2024-07-30T12:57:12.000+02:00"
      },
      {
        "posId": "CPOS0031234568",
        "terminalId": "POS-2-REC",
        "firstPosDateTime": "2024-07-30T09:40:12.000+02:00",
        "lastPosDateTime": "2024-07-30T09:45:12.000+02:00"
      }
    ],
    "fdmDevices": [
      {
        "fdmId": "FOD01000001",
        "firstFdmDateTime": "2024-07-30T06:30:12.000Z",
        "firstTotalCounter": 49,
        "lastFdmDateTime": "2024-07-30T10:57:18.000Z",
        "lastTotalCounter": 56
      }
    ],
    "turnover": {
      "transactions": [
        {
          "eventLabel": "N",
          "ticketCount": 4,
          "amount": 102.04
        },
        {
          "eventLabel": "P",
          "ticketCount": 3,
```

```
    "amount": 198
  }
],
"departments": [
  {
    "departmentId": "10",
    "departmentName": "Aperitifs",
    "amount": 24
  },
  {
    "departmentId": "12",
    "departmentName": "Soft Drinks",
    "amount": 3.04
  },
  {
    "departmentId": "15",
    "departmentName": "Wines",
    "amount": 35
  },
  {
    "departmentId": "22",
    "departmentName": "Main Dishes",
    "amount": 40
  }
],
"vats": [
  {
    "label": "A",
    "rate": 21,
    "taxableAmount": 51.27,
    "vatAmount": 10.77,
    "totalAmount": 62.04
  },
  {
    "label": "B",
    "rate": 12,
    "taxableAmount": 35.71,
    "vatAmount": 4.29,
    "totalAmount": 40
  }
],
"payments": [
  {
    "id": "1",
    "name": "Contant",
    "type": "CASH",
    "amount": [
      {
        "type": "PAYMENT",
        "normalAmount": 99,
        "negativeCorrections": 0,
        "positiveCorrections": 0,
        "correctionsCount": 0,
        "totalAmount": 99,
        "normalForeign": null,
```

```
            "correctionsForeign": null
          }
        ]
      },
      {
        "id": "2",
        "name": "Mastercard",
        "type": "CARD_CREDIT",
        "amount": [
          {
            "type": "PAYMENT",
            "normalAmount": 3.04,
            "negativeCorrections": 0,
            "positiveCorrections": 0,
            "correctionsCount": 0,
            "totalAmount": 3.04
          }
        ]
      }
    ],
    "drawersOpenCount": 0,
    "negQuantities": [
      {
        "negQuantityReason": "REFUND",
        "negQuantityCount": 2,
        "ticketCount": 1,
        "amount": -52
      }
    ],
    "priceChanges": [],
    "invoices": []
  }
 }
}
```


Response FDM :


```
{
  "data": {
    "signReportTurnoverX": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 208,
      "posDateTime": "2024-07-30T13:00:35.000+02:00",
      "terminalId": "POS-1-BAR",
      "deviceId": "F0:1D:3D:CC:13:C8",
      "eventOperation": "REPORT_TURNOVER_X",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-30T11:00:38.000Z",
        "eventLabel": "R",
        "eventCounter": 3,
        "totalCounter": 57
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
```

```
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/reports_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m180_turnover_x_of_the_scenario_day` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - la journée du scénario (InputReport T1–T8) est jouée par les vraies commandes, puis le rapport est construit par pili-pos à partir de son archive fiscale et comparé à la fixture par `crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m180_turnover_x_of_the_scenario_day`.

Requête du rapport (`evidence/uc/M180_event9_R_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "dc48bff8-d53e-4af1-8855-907a9f2765d1",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "fdmDevices": [
    {
      "fdmId": "FOD01000001",
      "firstFdmDateTime": "2024-07-30T06:30:19.000Z",
      "firstTotalCounter": 49,
      "lastFdmDateTime": "2024-07-30T10:57:18.000Z",
      "lastTotalCounter": 56
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:53.952+02:00",
  "posDevices": [
    {
      "firstPosDateTime": "2026-10-18T06:03:53.913+02:00",
      "lastPosDateTime": "2026-10-18T06:03:53.946+02:00",
      "posId": "CPOS0031234567",
      "terminalId": "POS-1-BAR"
    }
  ],
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "turnover": {
    "departments": [
      {
        "amount": 24.0,
        "departmentId": "10",
        "departmentName": "Aperitifs"
      },
      {
        "amount": 3.04,
        "departmentId": "12",
        "departmentName": "Soft Drinks"
```

```
    },
    {
      "amount": 35.0,
      "departmentId": "15",
      "departmentName": "Wines"
    },
    {
      "amount": 40.0,
      "departmentId": "22",
      "departmentName": "Main Dishes"
    }
  ],
  "drawersOpenCount": 0,
  "invoices": [],
  "negQuantities": [
    {
      "amount": -52.0,
      "negQuantityCount": 2,
      "negQuantityReason": "REFUND",
      "ticketCount": 1
    }
  ],
  "payments": [
    {
      "amount": [
        {
          "correctionsCount": 0,
          "negativeCorrections": 0.0,
          "normalAmount": 99.0,
          "positiveCorrections": 0.0,
          "totalAmount": 99.0,
          "type": "PAYMENT"
        }
      ],
      "id": "1",
      "name": "Cash",
      "type": "CASH"
    },
    {
      "amount": [
        {
          "correctionsCount": 0,
          "negativeCorrections": 0.0,
          "normalAmount": 3.04,
          "positiveCorrections": 0.0,
          "totalAmount": 3.04,
          "type": "PAYMENT"
        }
      ],
      "id": "2",
      "name": "Card",
      "type": "CARD_DEBIT"
    }
  ],
  "priceChanges": [],
```

```
    "transactions": [
      {
        "amount": 102.04,
        "eventLabel": "N",
        "ticketCount": 4
      },
      {
        "amount": 250.0,
        "eventLabel": "P",
        "ticketCount": 4
      }
    ],
    "vats": [
      {
        "label": "A",
        "rate": 21.0,
        "taxableAmount": 51.27,
        "totalAmount": 62.04,
        "vatAmount": 10.77
      },
      {
        "label": "B",
        "rate": 12.0,
        "taxableAmount": 35.71,
        "totalAmount": 40.0,
        "vatAmount": 4.29
      }
    ]
  },
  "vatNo": "BE0000000097"
}
```


Events de la journée, archivés de même : `evidence/uc/M180_event1_P_request.json`, `evidence/uc/M180_event2_N_r` `evidence/uc/M180_event3_N_request.json`, `evidence/uc/M180_event4_N_request.json`, `evidence/uc/M180_event5_P_request.json`, `evidence/uc/M180_event6_P_request.json`, `evidence/uc/M180_event7_P_request.json`, `evidence/uc/M180_event8_N_request.json`.

Comparés exactement : nombre d'events et montants par label, départements, TVA (taux, base, TVA, total), paiements, quantités négatives, ouvertures de tiroir, totaux, paiements et présences par utilisateur. Écarts assumés :

Champ	Fixture	pili-pos	Motif
`transactions[P]`	3 tickets, 198 €	4 tickets, 250 €	chaque event P de la période est compté (§2.2.2 `ticketCount`) : T1, T5, T6 et T7 (AM du 29/04/2024, art. 29, t)
`payments[].id` / `name`	identifiants et libellés du scénario	identifiants et libellés de pili-pos	propres à chaque POS
`payments[].type` (carte)	`CARD_CREDIT`	`CARD_DEBIT`	l'opérateur ne distingue pas carte de débit et de crédit
`departments[].name`	libellés du scénario	libellés français du catalogue	propres au catalogue
`posDevices` / `fdmDevices`	appareils du scénario	ce terminal et le FDM de test	propres à l'installation

M181_signReportTurnoverZ

Note de déclaration : le POS pili-pos n'émet jamais le Z financier seul : la clôture de journée émet toujours le **Z combiné M184** (`signReportTurnoverZ` + `signReportUserZ` dans un même document GraphQL, deux `posFiscalTicketNo` séquentiels) - voir la section M184 ci-dessous. La fixture M181 est validée en *parsing* et sa forme `signReportTurnoverZ` est exactement celle émise dans la première moitié du document M184. Couverture e2e applicative du Z : `m184_combined_z_of_the_scenario_day` dans `crates/pos-engine/tests/spf_e2e_report_fixtures.rs` (journée du scénario comparée à la fixture) et `m184_*` dans `crates/pos-engine/tests/spf_e2e_reports.rs`.

Mutation invoquee : `signReportTurnoverZ`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	NL	OK
`posId`	CPOS0031234567	OK
`posFiscalTicketNo`	209	OK
`posDateTime`	2024-07-30T13:05:35.000+02:00	OK
`terminalId`	POS-1-BAR	OK
`deviceId`	F0:1D:3D:CC:13:C8	OK
`vatNo`	BE0000000097	OK
`estNo`	2000000042	OK
`posSwVersion`	1.8.3	OK
`bookingPeriodId`	dffcd829-a0e5-41ca-a0ae-9eb887f95637	OK
`bookingDate`	2024-07-30	OK
`ticketMedium`	PAPER	OK
`employeeId`	75061189702	OK
`fdmRef.fdmId`	FOD01000001	OK
`fdmRef.eventLabel`	R	OK
`fdmRef.eventCounter`	4	OK
`fdmRef.totalCounter`	58	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 209,
    "posDateTime": "2024-07-30T13:05:35.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "F0:1D:3D:CC:13:C8",
    "terminalId": "POS-1-BAR",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-30",
    "ticketMedium": "PAPER",
    "employeeId": "75061189702",
    "reportNo": 2,
    "reportBookingDate": "2024-07-30",
    "posDevices": [
      {
        "posId": "CPOS0031234567",
        "terminalId": "TER-1-BAR",
```

```
      "firstPosDateTime": "2024-07-30T08:30:12.000+02:00",
      "lastPosDateTime": "2024-07-30T08:40:12.000+02:00"
    },
    {
      "posId": "CPOS0031234567",
      "terminalId": "TER-2-DIN",
      "firstPosDateTime": "2024-07-30T11:45:12.000+02:00",
      "lastPosDateTime": "2024-07-30T12:57:12.000+02:00"
    },
    {
      "posId": "CPOS0031234567",
      "terminalId": "POS-1-BAR",
      "firstPosDateTime": "2024-07-30T13:00:35.000+02:00",
      "lastPosDateTime": "2024-07-30T13:00:35.000+02:00"
    },
    {
      "posId": "CPOS0031234568",
      "terminalId": "POS-2-REC",
      "firstPosDateTime": "2024-07-30T09:40:12.000+02:00",
      "lastPosDateTime": "2024-07-30T09:45:12.000+02:00"
    }
  ],
  "fdmDevices": [
    {
      "fdmId": "FOD01000001",
      "firstFdmDateTime": "2024-07-30T06:30:12.000Z",
      "firstTotalCounter": 49,
      "lastFdmDateTime": "2024-07-30T11:00:38.000Z",
      "lastTotalCounter": 57
    }
  ],
  "turnover": {
    "transactions": [
      {
        "eventLabel": "N",
        "ticketCount": 4,
        "amount": 102.04
      },
      {
        "eventLabel": "P",
        "ticketCount": 3,
        "amount": 198
      }
    ],
    "departments": [
      {
        "departmentId": "10",
        "departmentName": "Aperitifs",
        "amount": 24
      },
      {
        "departmentId": "12",
        "departmentName": "Soft Drinks",
        "amount": 3.04
      },
```

```
    {
      "departmentId": "15",
      "departmentName": "Wines",
      "amount": 35
    },
    {
      "departmentId": "22",
      "departmentName": "Main Dishes",
      "amount": 40
    }
  ],
  "vats": [
    {
      "label": "A",
      "rate": 21,
      "taxableAmount": 51.27,
      "vatAmount": 10.77,
      "totalAmount": 62.04
    },
    {
      "label": "B",
      "rate": 12,
      "taxableAmount": 35.71,
      "vatAmount": 4.29,
      "totalAmount": 40
    }
  ],
  "payments": [
    {
      "id": "1",
      "name": "Contant",
      "type": "CASH",
      "amount": [
        {
          "type": "PAYMENT",
          "normalAmount": 99,
          "negativeCorrections": 0,
          "positiveCorrections": 0,
          "correctionsCount": 0,
          "totalAmount": 99,
          "normalForeign": null,
          "correctionsForeign": null
        }
      ]
    },
    {
      "id": "2",
      "name": "Mastercard",
      "type": "CARD_CREDIT",
      "amount": [
        {
          "type": "PAYMENT",
          "normalAmount": 3.04,
          "negativeCorrections": 0,
          "positiveCorrections": 0,
```

```
            "correctionsCount": 0,
            "totalAmount": 3.04
          }
        ]
      }
    ],
    "drawersOpenCount": 0,
    "negQuantities": [
      {
        "negQuantityReason": "REFUND",
        "negQuantityCount": 2,
        "ticketCount": 1,
        "amount": -52
      }
    ],
    "priceChanges": [],
    "invoices": []
  }
 }
}
```


Response FDM :


```
{
  "data": {
    "signReportTurnoverZ": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 209,
      "posDateTime": "2024-07-30T13:05:35.000+02:00",
      "terminalId": "POS-1-BAR",
      "deviceId": "F0:1D:3D:CC:13:C8",
      "eventOperation": "REPORT_TURNOVER_Z",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-30T11:05:41.000Z",
        "eventLabel": "R",
        "eventCounter": 4,
        "totalCounter": 58
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/reports_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m184_combined_z_of_the_scenario_day` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - cet UC est signé par le Z combiné de M184 (mutation `signReportTurnoverZ`), rejoué par `crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m184_combined_z_of_the_scenario_day` : voir `evidence/uc/M184_event9_R_request.json`, `evidence/uc/M184_event10_R_request.json` et la section M184.

M182_signReportUserX

Note de déclaration : implémenté par la commande `generate_user_x_report` (`crates/pos-engine/src/commands/rep` `do_generate_user_x_report`). Couverture e2e applicative : `m182_user_x_of_the_scenario_day` dans `crates/pos-engine/tests/spf_e2e_report_fixtures.rs` (journée du scénario comparée à la fixture) et `m182_generate_user_x_report_archives_r_event_with_user_block` dans `crates/pos-engine/tests/spf_e2e_reports.rs` ; preuves dans `evidence/uc/M182_*.json` (journée T1–T8 puis l'événement R).

Mutation invoquee : `signReportUserX`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	`NL`	OK
`posId`	`CPOS0031234567`	OK
`posFiscalTicketNo`	`210`	OK
`posDateTime`	`2024-07-30T13:10:35.000+02:00`	OK
`terminalId`	`POS-1-BAR`	OK
`deviceId`	`F0:1D:3D:CC:13:C8`	OK
`vatNo`	`BE0000000097`	OK
`estNo`	`2000000042`	OK
`posSwVersion`	`1.8.3`	OK
`bookingPeriodId`	`dffcd829-a0e5-41ca-a0ae-9eb887f95637`	OK
`bookingDate`	`2024-07-30`	OK
`ticketMedium`	`PAPER`	OK
`employeeId`	`75061189702`	OK
`fdmRef.fdmId`	`FOD01000001`	OK
`fdmRef.eventLabel`	`R`	OK
`fdmRef.eventCounter`	`5`	OK
`fdmRef.totalCounter`	`59`	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 210,
    "posDateTime": "2024-07-30T13:10:35.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "F0:1D:3D:CC:13:C8",
    "terminalId": "POS-1-BAR",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-30",
    "ticketMedium": "PAPER",
    "employeeId": "75061189702",
    "posDevices": [
      {
        "posId": "CPOS0031234567",
        "terminalId": "TER-1-BAR",
        "firstPosDateTime": "2024-07-30T08:30:12.000+02:00",
        "lastPosDateTime": "2024-07-30T08:40:12.000+02:00"
      },
```

```
    {
      "posId": "CPOS0031234567",
      "terminalId": "TER-2-DIN",
      "firstPosDateTime": "2024-07-30T11:45:12.000+02:00",
      "lastPosDateTime": "2024-07-30T12:57:12.000+02:00"
    },
    {
      "posId": "CPOS0031234567",
      "terminalId": "POS-1-BAR",
      "firstPosDateTime": "2024-07-30T13:00:35.000+02:00",
      "lastPosDateTime": "2024-07-30T13:05:35.000+02:00"
    },
    {
      "posId": "CPOS0031234568",
      "terminalId": "POS-2-REC",
      "firstPosDateTime": "2024-07-30T09:40:12.000+02:00",
      "lastPosDateTime": "2024-07-30T09:45:12.000+02:00"
    }
  ],
  "fdmDevices": [
    {
      "fdmId": "FOD01000001",
      "firstFdmDateTime": "2024-04-22T06:30:12.000Z",
      "firstTotalCounter": 49,
      "lastFdmDateTime": "2024-07-30T11:05:41.000Z",
      "lastTotalCounter": 58
    }
  ],
  "users": [
    {
      "employeeId": "75061189702",
      "totalAmount": 0,
      "firstPosDateTime": "2024-07-30T08:30:12.000+02:00",
      "lastPosDateTime": "2024-07-30T09:40:12.000+02:00",
      "socialEvents": [],
      "payments": [
        {
          "id": "1",
          "name": "CONTANT",
          "type": "CASH",
          "amount": [
            {
              "type": "PAYMENT",
              "normalAmount": 0,
              "negativeCorrections": 0,
              "positiveCorrections": 0,
              "correctionsCount": 0,
              "totalAmount": 0
            }
          ]
        }
      ]
    },
    {
      "employeeId": "84022899837",
```

```
        "totalAmount": 102.04,
        "firstPosDateTime": "2024-07-30T09:45:12.000+02:00",
        "lastPosDateTime": "2024-07-30T12:57:12.000+02:00",
        "socialEvents": [],
        "payments": [
          {
            "id": "1",
            "name": "CONTANT",
            "type": "CASH",
            "amount": [
              {
                "type": "PAYMENT",
                "normalAmount": 99,
                "negativeCorrections": 0,
                "positiveCorrections": 0,
                "correctionsCount": 0,
                "totalAmount": 99
              }
            ]
          },
          {
            "id": "2",
            "name": "Master Card",
            "type": "CARD_CREDIT",
            "amount": [
              {
                "type": "PAYMENT",
                "normalAmount": 3.04,
                "negativeCorrections": 0,
                "positiveCorrections": 0,
                "correctionsCount": 0,
                "totalAmount": 3.04
              }
            ]
          }
        ]
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signReportUserX": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 210,
      "posDateTime": "2024-07-30T13:10:35.000+02:00",
      "terminalId": "POS-1-BAR",
      "deviceId": "F0:1D:3D:CC:13:C8",
      "eventOperation": "REPORT_USER_X",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-30T11:10:42.000Z",
        "eventLabel": "R",
```

```
        "eventCounter": 5,
        "totalCounter": 59
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/reports_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m182_user_x_of_the_scenario_day` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - la journée du scénario (InputReport T1–T8) est jouée par les vraies commandes, puis le rapport est construit par pili-pos à partir de son archive fiscale et comparé à la fixture par `crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m182_user_x_of_the_scenario_day`.

Requête du rapport (`evidence/uc/M182_event9_R_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "99da09ad-a9e7-4a41-a5f9-3b5a9e061ba2",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "fdmDevices": [
    {
      "fdmId": "FOD01000001",
      "firstFdmDateTime": "2024-07-30T06:30:19.000Z",
      "firstTotalCounter": 49,
      "lastFdmDateTime": "2024-07-30T10:57:18.000Z",
      "lastTotalCounter": 56
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:53.943+02:00",
  "posDevices": [
    {
      "firstPosDateTime": "2026-10-18T06:03:53.902+02:00",
      "lastPosDateTime": "2026-10-18T06:03:53.937+02:00",
      "posId": "CPOS0031234567",
      "terminalId": "POS-1-BAR"
    }
  ],
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "users": [
    {
      "employeeId": "75061189702",
      "firstPosDateTime": "2026-10-18T06:03:53.902+02:00",
      "lastPosDateTime": "2026-10-18T06:03:53.913+02:00",
```

```
  "payments": [
    {
      "amount": [
        {
          "correctionsCount": 0,
          "negativeCorrections": 0.0,
          "normalAmount": 0.0,
          "positiveCorrections": 0.0,
          "totalAmount": 0.0,
          "type": "PAYMENT"
        }
      ],
      "id": "1",
      "name": "Cash",
      "type": "CASH"
    }
  ],
  "socialEvents": [],
  "totalAmount": 0.0
},
{
  "employeeId": "84022899837",
  "firstPosDateTime": "2026-10-18T06:03:53.919+02:00",
  "lastPosDateTime": "2026-10-18T06:03:53.937+02:00",
  "payments": [
    {
      "amount": [
        {
          "correctionsCount": 0,
          "negativeCorrections": 0.0,
          "normalAmount": 99.0,
          "positiveCorrections": 0.0,
          "totalAmount": 99.0,
          "type": "PAYMENT"
        }
      ],
      "id": "1",
      "name": "Cash",
      "type": "CASH"
    },
    {
      "amount": [
        {
          "correctionsCount": 0,
          "negativeCorrections": 0.0,
          "normalAmount": 3.04,
          "positiveCorrections": 0.0,
          "totalAmount": 3.04,
          "type": "PAYMENT"
        }
      ],
      "id": "2",
      "name": "Card",
      "type": "CARD_DEBIT"
    }
```

```
      ],
      "socialEvents": [],
      "totalAmount": 102.04
    }
  ],
  "vatNo": "BE0000000097"
}
```


Events de la journée, archivés de même : `evidence/uc/M182_event1_P_request.json`, `evidence/uc/M182_event2_N_request.json`, `evidence/uc/M182_event3_N_request.json`, `evidence/uc/M182_event4_N_request.json`, `evidence/uc/M182_event5_P_request.json`, `evidence/uc/M182_event6_P_request.json`, `evidence/uc/M182_event7_P_request.json`, `evidence/uc/M182_event8_N_request.json`.

Comparés exactement : nombre d'events et montants par label, départements, TVA (taux, base, TVA, total), paiements, quantités négatives, ouvertures de tiroir, totaux, paiements et présences par utilisateur. Écarts assumés :

Champ	Fixture	pili-pos	Motif
`transactions[P]`	3 tickets, 198 €	4 tickets, 250 €	chaque event P de la période est compté (§2.2.2 `ticketCount`) : T1, T5, T6 et T7 (AM du 29/04/2024, art. 29, t)
`payments[].id` / `name`	identifiants et libellés du scénario	identifiants et libellés de pili-pos	propres à chaque POS
`payments[].type` (carte)	`CARD_CREDIT`	`CARD_DEBIT`	l'opérateur ne distingue pas carte de débit et de crédit
`departments[].name`	libellés du scénario	libellés français du catalogue	propres au catalogue
`posDevices` / `fdmDevices`	appareils du scénario	ce terminal et le FDM de test	propres à l'installation

M183_signReportUserZ

Note de déclaration : comme pour M181, le POS n'émet jamais le Z utilisateurs seul - il est toujours émis dans le document combiné M184 (seconde moitié). La fixture M183 est validée en *parsing* et sa forme `signReportUserZ` est exactement celle émise dans le document M184.

Mutation invoquee : `signReportUserZ`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`language`	`NL`	OK
`posId`	`CPOS0031234567`	OK
`posFiscalTicketNo`	`211`	OK
`posDateTime`	`2024-07-30T13:15:35.000+02:00`	OK
`terminalId`	`POS-1-BAR`	OK
`deviceId`	`F0:1D:3D:CC:13:C8`	OK
`vatNo`	`BE0000000097`	OK
`estNo`	`2000000042`	OK
`posSwVersion`	`1.8.3`	OK
`bookingPeriodId`	`dffcd829-a0e5-41ca-a0ae-9eb887f95637`	OK
`bookingDate`	`2024-07-30`	OK

Champ	Valeur fixture	Statut
ticketMedium	PAPER	OK
employeeId	75061189702	OK
fdmRef.fdmId	FOD01000001	OK
fdmRef.eventLabel	R	OK
fdmRef.eventCounter	6	OK
fdmRef.totalCounter	60	OK

Request (Variables.json) :


```
{
  "data": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 211,
    "posDateTime": "2024-07-30T13:15:35.000+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "F0:1D:3D:CC:13:C8",
    "terminalId": "POS-1-BAR",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-30",
    "ticketMedium": "PAPER",
    "employeeId": "75061189702",
    "reportBookingDate": "2024-07-30",
    "reportNo": 2,
    "posDevices": [
      {
        "posId": "CPOS0031234567",
        "terminalId": "TER-1-BAR",
        "firstPosDateTime": "2024-07-30T08:30:12.000+02:00",
        "lastPosDateTime": "2024-07-30T08:40:12.000+02:00"
      },
      {
        "posId": "CPOS0031234567",
        "terminalId": "TER-2-DIN",
        "firstPosDateTime": "2024-07-30T11:45:12.000+02:00",
        "lastPosDateTime": "2024-07-30T12:57:12.000+02:00"
      },
      {
        "posId": "CPOS0031234567",
        "terminalId": "POS-1-BAR",
        "firstPosDateTime": "2024-07-30T13:00:35.000+02:00",
        "lastPosDateTime": "2024-07-30T13:10:35.000+02:00"
      },
      {
        "posId": "CPOS0031234568",
        "terminalId": "POS-2-REC",
        "firstPosDateTime": "2024-07-30T09:40:12.000+02:00",
        "lastPosDateTime": "2024-07-30T09:45:12.000+02:00"
      }
    ],
    "fdmDevices": [
      {
```

```
    "fdmId": "FOD01000001",
    "firstFdmDateTime": "2024-04-22T06:30:12.000Z",
    "firstTotalCounter": 49,
    "lastFdmDateTime": "2024-07-30T11:10:42.000Z",
    "lastTotalCounter": 59
  }
],
"users": [
  {
    "employeeId": "75061189702",
    "totalAmount": 0,
    "firstPosDateTime": "2024-07-30T08:30:12.000+02:00",
    "lastPosDateTime": "2024-07-30T09:40:12.000+02:00",
    "socialEvents": [],
    "payments": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "amount": [
          {
            "type": "PAYMENT",
            "normalAmount": 0,
            "negativeCorrections": 0,
            "positiveCorrections": 0,
            "correctionsCount": 0,
            "totalAmount": 0
          }
        ]
      }
    ]
  },
  {
    "employeeId": "84022899837",
    "totalAmount": 102.04,
    "firstPosDateTime": "2024-07-30T09:45:12.000+02:00",
    "lastPosDateTime": "2024-07-30T12:57:12.000+02:00",
    "socialEvents": [],
    "payments": [
      {
        "id": "1",
        "name": "CONTANT",
        "type": "CASH",
        "amount": [
          {
            "type": "PAYMENT",
            "normalAmount": 99,
            "negativeCorrections": 0,
            "positiveCorrections": 0,
            "correctionsCount": 0,
            "totalAmount": 99
          }
        ]
      },
      {
```

```
            "id": "2",
            "name": "Master Card",
            "type": "CARD_CREDIT",
            "amount": [
              {
                "type": "PAYMENT",
                "normalAmount": 3.04,
                "negativeCorrections": 0,
                "positiveCorrections": 0,
                "correctionsCount": 0,
                "totalAmount": 3.04
              }
            ]
          }
        ]
      }
    ]
  }
}
```


Response FDM :


```
{
  "data": {
    "signReportUserZ": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 211,
      "posDateTime": "2024-07-30T13:15:35.000+02:00",
      "terminalId": "POS-1-BAR",
      "deviceId": "F0:1D:3D:CC:13:C8",
      "eventOperation": "REPORT_USER_Z",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-30T11:15:41.000Z",
        "eventLabel": "R",
        "eventCounter": 6,
        "totalCounter": 60
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/reports_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m184_combined_z_of_the_scenario_day` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - cet UC est signé par le Z combiné de M184 (mutation `signReportUserZ`), rejoué par `crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m184_combined_z_of_the_scenario_day` : voir `evidence/uc/M184_event9_R_request.json`, `evidence/uc/M184_event10_R_request.json` et la section M184.

M184_signReportTurnoverZUserZ

Note de déclaration : c'est CE mode d'émission du Z que le POS implémente et déclare : `do_generate_z_report` (`crates/pos-engine/src/commands/report.rs`) envoie `signReportTurnoverZ + signReportUserZ` dans **un seul document GraphQL** (`FdmClient::sign_report_turnover_z_user_z`), alloue deux `pos-FiscalTicketNo` séquentiels et archive les deux événements R dans une même transaction SQLite. Couverture e2e applicative : `m184_generate_z_report_signs_both_reports_in_one_operation` et `z_report_starts_new_booking_period` dans `crates/pos-engine/tests/spf_e2e_reports.rs`, et `m184_combined_z_of_the_scenario_day` dans `crates/pos-engine/tests/spf_e2e_report_fixtures.rs` (journée du scénario comparée à la fixture) ; preuves dans `evidence/uc/M184_*.json` (journée T1–T8 puis les 2 événements R).

Mutation invoquee : `signReportTurnoverZ`

Champs a verifier (manuel SCE 2.0 §IV) :

Champ	Valeur fixture	Statut
`fdmRef.fdmId`	`FOD01000001`	OK
`fdmRef.eventLabel`	R	OK
`fdmRef.eventCounter`	3	OK
`fdmRef.totalCounter`	47	OK

Request (Variables.json) :


```
{
  "turnover": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 209,
    "posDateTime": "2024-07-30T13:05:35.521+02:00",
    "posSwVersion": "1.8.3",
    "deviceId": "F0:1D:3D:CC:13:C8",
    "terminalId": "POS-1-BAR",
    "bookingPeriodId": "dffcd829-a0e5-41ca-a0ae-9eb887f95637",
    "bookingDate": "2024-07-30",
    "ticketMedium": "PAPER",
    "employeeId": "75061189702",
    "reportNo": 2,
    "reportBookingDate": "2024-07-30",
    "posDevices": [
      {
        "posId": "CPOS0031234567",
        "terminalId": "TER-1-BAR",
        "firstPosDateTime": "2024-07-30T08:30:12+02:00",
        "lastPosDateTime": "2024-07-30T08:40:12+02:00"
      },
      {
        "posId": "CPOS0031234567",
        "terminalId": "TER-2-DIN",
        "firstPosDateTime": "2024-07-30T11:45:12+02:00",
        "lastPosDateTime": "2024-07-30T12:57:12+02:00"
      },
      {
        "posId": "CPOS0031234567",
```

```
      "terminalId": "POS-1-BAR",
      "firstPosDateTime": "2024-07-30T13:00:35+02:00",
      "lastPosDateTime": "2024-07-30T13:00:35+02:00"
    },
    {
      "posId": "CPOS0031234568",
      "terminalId": "POS-2-REC",
      "firstPosDateTime": "2024-07-30T09:40:12+02:00",
      "lastPosDateTime": "2024-07-30T09:45:12+02:00"
    }
  ],
  "fdmDevices": [
    {
      "fdmId": "FOD01000001",
      "firstFdmDateTime": "2024-07-30T06:30:12Z",
      "firstTotalCounter": 49,
      "lastFdmDateTime": "2024-07-30T11:00:38Z",
      "lastTotalCounter": 57
    }
  ],
  "turnover": {
    "transactions": [
      {
        "eventLabel": "N",
        "ticketCount": 4,
        "amount": 102.04
      },
      {
        "eventLabel": "P",
        "ticketCount": 3,
        "amount": 198
      }
    ],
    "departments": [
      {
        "departmentId": "10",
        "departmentName": "Aperitifs",
        "amount": 24
      },
      {
        "departmentId": "12",
        "departmentName": "Soft Drinks",
        "amount": 3.04
      },
      {
        "departmentId": "15",
        "departmentName": "Wines",
        "amount": 35
      },
      {
        "departmentId": "22",
        "departmentName": "Main Dishes",
        "amount": 40
      }
    ],
```

```
"vats": [
  {
    "label": "A",
    "rate": 21,
    "taxableAmount": 51.27,
    "vatAmount": 10.77,
    "totalAmount": 62.04
  },
  {
    "label": "B",
    "rate": 12,
    "taxableAmount": 35.71,
    "vatAmount": 4.29,
    "totalAmount": 40
  }
],
"payments": [
  {
    "id": "1",
    "name": "Contant",
    "type": "CASH",
    "amount": [
      {
        "type": "PAYMENT",
        "normalAmount": 99,
        "negativeCorrections": 0,
        "positiveCorrections": 0,
        "correctionsCount": 0,
        "totalAmount": 99,
        "normalForeign": null,
        "correctionsForeign": null
      }
    ]
  },
  {
    "id": "2",
    "name": "Mastercard",
    "type": "CARD_CREDIT",
    "amount": [
      {
        "type": "PAYMENT",
        "normalAmount": 3.04,
        "negativeCorrections": 0,
        "positiveCorrections": 0,
        "correctionsCount": 0,
        "totalAmount": 3.04
      }
    ]
  }
],
"drawersOpenCount": 0,
"negQuantities": [
  {
    "negQuantityReason": "REFUND",
    "negQuantityCount": 2,
```

```
          "ticketCount": 1,
          "amount": -52
        }
      ],
      "priceChanges": [],
      "invoices": []
    }
  },
  "user": {
    "language": "NL",
    "vatNo": "BE0000000097",
    "estNo": "2000000042",
    "posId": "CPOS0031234567",
    "posFiscalTicketNo": 210,
    "posDateTime": "202
... (tronque, voir fixture source)
```


Response FDM :


```
{
  "data": {
    "signReportTurnoverZ": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 209,
      "posDateTime": "2024-07-30T13:05:35.000+02:00",
      "terminalId": "POS-1-BAR",
      "deviceId": "F0:1D:3D:CC:13:C8",
      "eventOperation": "REPORT_TURNOVER_Z",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-30T11:05:43.000Z",
        "eventLabel": "R",
        "eventCounter": 3,
        "totalCounter": 47
      },
      "fdmSwVersion": "1.2.0",
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    },
    "signReportUserZ": {
      "posId": "CPOS0031234567",
      "posFiscalTicketNo": 210,
      "posDateTime": "2024-07-30T13:05:35.000+02:00",
      "terminalId": "POS-1-BAR",
      "deviceId": "F0:1D:3D:CC:13:C8",
      "eventOperation": "REPORT_USER_Z",
      "fdmRef": {
        "fdmId": "FOD01000001",
        "fdmDateTime": "2024-07-30T11:05:43.000Z",
        "eventLabel": "R",
        "eventCounter": 4,
        "totalCounter": 48
      },
      "fdmSwVersion": "1.2.0",
```

```
      "bufferCapacityUsed": 0.25,
      "digitalSignature": "c7uwqjLZ+Wk+R8swA/KHilWp4qCPK4u1L6TGxuCOJbf7bY32Ra2G8NiI+TO9lFGNFTfU1U
      "warnings": [],
      "informations": []
    }
  }
}
```


Validation automatisee : `crates/pos-fiscal/tests/reports_fixtures.rs` (lecture de la fixture) et `crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m184_combined_z_of_the_scenario_day` (rejeu applicatif, build `cert` ; voir « Preuve pili-pos » ci-dessous).

Preuve pili-pos (build `cert`) - la journée du scénario (InputReport T1–T8) est jouée par les vraies commandes, puis le rapport est construit par pili-pos à partir de son archive fiscale et comparé à la fixture par `crates/pos-engine/tests/spf_e2e_report_fixtures.rs::m184_combined_z_of_the_scenario_day`.

Requête du rapport (`evidence/uc/M184_event9_R_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "fc6b852e-7820-4c5d-bf15-61b0ec2f45bf",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "fdmDevices": [
    {
      "fdmId": "FOD01000001",
      "firstFdmDateTime": "2024-07-30T06:30:19.000Z",
      "firstTotalCounter": 49,
      "lastFdmDateTime": "2024-07-30T10:57:18.000Z",
      "lastTotalCounter": 56
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:53.966+02:00",
  "posDevices": [
    {
      "firstPosDateTime": "2026-10-18T06:03:53.920+02:00",
      "lastPosDateTime": "2026-10-18T06:03:53.960+02:00",
      "posId": "CPOS0031234567",
      "terminalId": "POS-1-BAR"
    }
  ],
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "reportBookingDate": "2026-10-18",
  "reportNo": 1,
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "turnover": {
    "departments": [
      {
        "amount": 24.0,
        "departmentId": "10",
        "departmentName": "Aperitifs"
      },
```

```
    {
      "amount": 3.04,
      "departmentId": "12",
      "departmentName": "Soft Drinks"
    },
    {
      "amount": 35.0,
      "departmentId": "15",
      "departmentName": "Wines"
    },
    {
      "amount": 40.0,
      "departmentId": "22",
      "departmentName": "Main Dishes"
    }
  ],
  "drawersOpenCount": 0,
  "invoices": [],
  "negQuantities": [
    {
      "amount": -52.0,
      "negQuantityCount": 2,
      "negQuantityReason": "REFUND",
      "ticketCount": 1
    }
  ],
  "payments": [
    {
      "amount": [
        {
          "correctionsCount": 0,
          "negativeCorrections": 0.0,
          "normalAmount": 99.0,
          "positiveCorrections": 0.0,
          "totalAmount": 99.0,
          "type": "PAYMENT"
        }
      ],
      "id": "1",
      "name": "Cash",
      "type": "CASH"
    },
    {
      "amount": [
        {
          "correctionsCount": 0,
          "negativeCorrections": 0.0,
          "normalAmount": 3.04,
          "positiveCorrections": 0.0,
          "totalAmount": 3.04,
          "type": "PAYMENT"
        }
      ],
      "id": "2",
      "name": "Card",
```

```
        "type": "CARD_DEBIT"
      }
    ],
    "priceChanges": [],
    "transactions": [
      {
        "amount": 102.04,
        "eventLabel": "N",
        "ticketCount": 4
      },
      {
        "amount": 250.0,
        "eventLabel": "P",
        "ticketCount": 4
      }
    ],
    "vats": [
      {
        "label": "A",
        "rate": 21.0,
        "taxableAmount": 51.27,
        "totalAmount": 62.04,
        "vatAmount": 10.77
      },
      {
        "label": "B",
        "rate": 12.0,
        "taxableAmount": 35.71,
        "totalAmount": 40.0,
        "vatAmount": 4.29
      }
    ]
  },
  "vatNo": "BE0000000097"
}
```


Requête du rapport (`evidence/uc/M184_event10_R_request.json`) :


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "fc6b852e-7820-4c5d-bf15-61b0ec2f45bf",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "75061189702",
  "estNo": "2000000042",
  "fdmDevices": [
    {
      "fdmId": "FOD01000001",
      "firstFdmDateTime": "2024-07-30T06:30:19.000Z",
      "firstTotalCounter": 49,
      "lastFdmDateTime": "2024-07-30T10:57:18.000Z",
      "lastTotalCounter": 56
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:53.966+02:00",
  "posDevices": [
```

```
    {
      "firstPosDateTime": "2026-10-18T06:03:53.920+02:00",
      "lastPosDateTime": "2026-10-18T06:03:53.960+02:00",
      "posId": "CPOS0031234567",
      "terminalId": "POS-1-BAR"
    }
  ],
  "posFiscalTicketNo": 2,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "reportBookingDate": "2026-10-18",
  "reportNo": 1,
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "users": [
    {
      "employeeId": "75061189702",
      "firstPosDateTime": "2026-10-18T06:03:53.920+02:00",
      "lastPosDateTime": "2026-10-18T06:03:53.929+02:00",
      "payments": [
        {
          "amount": [
            {
              "correctionsCount": 0,
              "negativeCorrections": 0.0,
              "normalAmount": 0.0,
              "positiveCorrections": 0.0,
              "totalAmount": 0.0,
              "type": "PAYMENT"
            }
          ],
          "id": "1",
          "name": "Cash",
          "type": "CASH"
        }
      ],
      "socialEvents": [],
      "totalAmount": 0.0
    },
    {
      "employeeId": "84022899837",
      "firstPosDateTime": "2026-10-18T06:03:53.936+02:00",
      "lastPosDateTime": "2026-10-18T06:03:53.960+02:00",
      "payments": [
        {
          "amount": [
            {
              "correctionsCount": 0,
              "negativeCorrections": 0.0,
              "normalAmount": 99.0,
              "positiveCorrections": 0.0,
              "totalAmount": 99.0,
              "type": "PAYMENT"
            }
          ],
```

```
          "id": "1",
          "name": "Cash",
          "type": "CASH"
        },
        {
          "amount": [
            {
              "correctionsCount": 0,
              "negativeCorrections": 0.0,
              "normalAmount": 3.04,
              "positiveCorrections": 0.0,
              "totalAmount": 3.04,
              "type": "PAYMENT"
            }
          ],
          "id": "2",
          "name": "Card",
          "type": "CARD_DEBIT"
        }
      ],
      "socialEvents": [],
      "totalAmount": 102.04
    }
  ],
  "vatNo": "BE0000000097"
}
```


Events de la journée, archivés de même : `evidence/uc/M184_event1_P_request.json`, `evidence/uc/M184_event2_N_r` `evidence/uc/M184_event3_N_request.json`, `evidence/uc/M184_event4_N_request.json`, `evidence/uc/M184_event5_P_request.json`, `evidence/uc/M184_event6_P_request.json`, `evidence/uc/M184_event7_P_request.json`, `evidence/uc/M184_event8_N_request.json`.

Comparés exactement : nombre d'events et montants par label, départements, TVA (taux, base, TVA, total), paiements, quantités négatives, ouvertures de tiroir, totaux, paiements et présences par utilisateur. Écarts assumés :

Champ	Fixture	pili-pos	Motif
`transactions[P]`	3 tickets, 198 €	4 tickets, 250 €	chaque event P de la période est compté (§2.2.2 `ticketCount`) : T1, T5, T6 et T7 (AM du 29/04/2024, art. 29, t)
`payments[].id` / `name`	identifiants et libellés du scénario	identifiants et libellés de pili-pos	propres à chaque POS
`payments[].type` (carte)	`CARD_CREDIT`	`CARD_DEBIT`	l'opérateur ne distingue pas carte de débit et de crédit
`departments[].name`	libellés du scénario	libellés français du catalogue	propres au catalogue
`posDevices` / `fdmDevices`	appareils du scénario	ce terminal et le FDM de test	propres à l'installation

Factures de l'Advanced Set (UC4121, UC4122)

Les deux UC de facture de l'Advanced Set v1.6 sont rejoués par `crates/pos-engine/tests/invoice_advanced.rs`, par la commande de facturation de l'écran, contre le module de contrôle simulé. Le scénario paie les ventes en crédit client ; ce moyen de paiement n'existe pas dans le produit, les ventes sont donc encaissées au comptant (la demande de facture ne dépend pas du moyen de paiement des ventes). Le test compare la requête archivée à la fixture (champs métier égaux, références des ventes comparées valeur par valeur à celles que le module de contrôle a répondues) et vérifie le format du numéro de facture.

UC4121_CustomerInvoice

Mutation invoquee : `signInvoice`

Scenario (fixture, en anglais) :

```
Sale of 2 carpaccio (unit price 20EUR) and 1 bottle of Chardonnay (unit price 40EUR), payment type CUST
Sale of 2 dry Martini (unit price 12EUR), payment type CUSTOMER CREDIT
Sale of 2 dame blanche (unit price 11,52EUR), payment type CUSTOMER CREDIT
End of the month: invoice with reference to tickets
```

Preuve pili-pos (build `cert`) - scénario rejoué par `crates/pos-engine/tests/invoice_advanced.rs::uc4121_an_invo` avec des ventes au comptant génériques à la place des étapes de vente de la fixture (le crédit client n'existe pas dans le produit ; la demande de facture n'en dépend pas). Requête **construite et archivée par pili-pos** (`evidence/uc/UC4121_event4_I_request.json`), réponse FDM archivée : `evidence/uc/UC4121_event4_I_response.`


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "9d9f66de-d519-48aa-b253-4724c280dfcb",
  "costCenter": {
    "id": "C1",
    "reference": "JANSSENS",
    "type": "CUSTOMER"
  },
  "customerVatNo": "BE0308357159",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "84022899837",
  "estNo": "2000000042",
  "fdmRefs": [
    {
      "eventCounter": 1,
      "eventLabel": "N",
      "fdmDateTime": "2026-10-18T04:03:46Z",
      "fdmId": "MCK00000001",
      "totalCounter": 1
    },
    {
      "eventCounter": 2,
      "eventLabel": "N",
      "fdmDateTime": "2026-10-18T04:03:46Z",
      "fdmId": "MCK00000001",
      "totalCounter": 2
    },
    {
      "eventCounter": 3,
      "eventLabel": "N",
      "fdmDateTime": "2026-10-18T04:03:46Z",
      "fdmId": "MCK00000001",
```

```
      "totalCounter": 3
    }
  ],
  "invoiceNo": "20260001",
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:47.002+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "vatNo": "BE0000000097"
}
```


Events préalables du scénario, archivés de même : `evidence/uc/UC4121_event1_N_request.json`, `evidence/uc/UC4121_event2_N_request.json`, `evidence/uc/UC4121_event3_N_request.json`.

Comparaison avec la fixture : champs métier identiques (`costCenter.type` et `costCenter.reference`, `customerVatNo`, libellés des `fdmRefs`) ; chaque `fdmRefs[i]` (`fdmId`, `fdmDateTime`, `eventLabel`, `eventCounter`, `totalCounter`) est comparé à la référence que le module de contrôle a répondue pour la vente correspondante, dans l'ordre ; numéro de facture contrôlé sur son format (année et séquence). Écarts voulus : numéro de facture, identifiant du centre de coût (`C` suivi du numéro du client) et compteurs, qui sont ceux de la session de test.

UC4122_Creditnote

Mutation invoquee : `signInvoice`

Scenario (fixture, en anglais) :

```
Refund of ticket, sale of 4 dame blanche (unit price 11.52EUR) had already been invoiced
Sale 1 of 4 dry Martini (unit price:12EUR)
Sale 2 of 4 burger of the chef (unit price: 28 EUR) and 4 Coca-Cola (unit price 3.04EUR)
Invoice of refund and sale 1 & 2
```

Le rejeu respecte la précondition : la vente de 46,08 EUR est d'abord facturée (première facture), puis remboursée par un événement N négatif ; la seconde facture reprend ce remboursement et les deux ventes (total 46,08 EUR en négatif, 48,00 EUR et 120,00 EUR en positif). La seconde tentative de facturer la vente d'origine est refusée (une vente, une facture).

Preuve pili-pos (build `cert`) - scénario rejoué par `crates/pos-engine/tests/invoice_advanced.rs::uc4122_an_invo` avec des ventes au comptant génériques à la place des étapes de vente de la fixture (le crédit client n'existe pas dans le produit ; la demande de facture n'en dépend pas). Requête **construite et archivée par pili-pos** (`evidence/uc/UC4122_event6_I_request.json`), réponse FDM archivée : `evidence/uc/UC4122_event6_I_response.`


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "298d4907-e23a-4851-8c50-be7af34a36b6",
  "costCenter": {
    "id": "C1",
    "reference": "AUDI",
    "type": "CUSTOMER"
  },
  "customerVatNo": "BE0308357159",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "84022899837",
  "estNo": "2000000042",
  "fdmRefs": [
    {
```

```
      "eventCounter": 2,
      "eventLabel": "N",
      "fdmDateTime": "2026-10-18T04:03:46Z",
      "fdmId": "MCK00000001",
      "totalCounter": 3
    },
    {
      "eventCounter": 3,
      "eventLabel": "N",
      "fdmDateTime": "2026-10-18T04:03:46Z",
      "fdmId": "MCK00000001",
      "totalCounter": 4
    },
    {
      "eventCounter": 4,
      "eventLabel": "N",
      "fdmDateTime": "2026-10-18T04:03:47Z",
      "fdmId": "MCK00000001",
      "totalCounter": 5
    }
  ],
  "invoiceNo": "20260002",
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:47.006+02:00",
  "posFiscalTicketNo": 2,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "vatNo": "BE0000000097"
}
```


Events préalables du scénario, archivés de même : `evidence/uc/UC4122_event1_N_request.json`, `evidence/uc/UC4122_event2_I_request.json`, `evidence/uc/UC4122_event3_N_request.json`, `evidence/uc/UC4122_event4_N_request.json`, `evidence/uc/UC4122_event5_N_request.json`.

Comparaison avec la fixture : champs métier identiques (`costCenter.type` et `costCenter.reference`, `customerVatNo`, libellés des `fdmRefs`) ; chaque `fdmRefs[i]` (`fdmId`, `fdmDateTime`, `eventLabel`, `eventCounter`, `totalCounter`) est comparé à la référence que le module de contrôle a répondue pour la vente correspondante, dans l'ordre ; numéro de facture contrôlé sur son format (année et séquence). Écarts voulus : numéro de facture, identifiant du centre de coût (`C` suivi du numéro du client) et compteurs, qui sont ceux de la session de test.

Arrondissement des methodes de paiement (manuel §IV)

Le manuel demande les tests de base 1 et 2 dans tous les cas, le test 3 si le POS n'arrondit que les paiements en especes, les tests 4 et 5 si tous les moyens de paiement sont arrondis. pili-pos n'arrondit **que les especes** (`crates/pos-fiscal/src/payment.rs` : un paiement par carte ou par cheque-repas est toujours exact, l'arrondi au 0,05 EUR porte sur le solde paye en especes, sur sa propre ligne `ROUNDING`). Sont donc presentees les cinq situations des tests de base 1 a 3 (les tests 4 et 5 ne s'appliquent pas). Produits du manuel : Dame Blanche 11,52 EUR et Tapas Variation 3,34 EUR, TVA B (12 %). Les situations sont rejouees par `crates/pos-engine/tests/spf_e2e_rounding.rs`, par les commandes d'encaissement, contre le FDM simule ; le test compare les lignes `financials` de la requete et le ticket imprime au tableau du manuel.

ROUND1_Arrondi_cash_Dame_Blanche

Test de base 1, situation 1. Vente directe de 1x Dame Blanche (11,52 EUR), payee uniquement en especes. Attendu : a payer 11,50 EUR, arrondi -0,02 EUR.

Preuve pili-pos (build `cert`) - situation rejouée par `crates/pos-engine/tests/spf_e2e_rounding.rs::round1_dame_b` par la commande que l'écran d'encaissement appelle. Requête **construite et archivée par pili-pos** (`evidence/uc/ROUND1_event1_N_request.json`) ; réponse FDM archivée : `evidence/uc/ROUND1_event1_N_response`


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "effc5cd6-efcb-4702-bf7e-de43bb35950f",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "12345678958",
  "estNo": "2000000042",
  "financials": [
    {
      "amount": 11.52,
      "amountType": "PAYMENT",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    },
    {
      "amount": -0.02,
      "amountType": "ROUNDING",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:55.006+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "transaction": {
    "transactionLines": [
      {
        "lineTotal": 11.52,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "1",
          "departmentName": "Desserts",
          "productId": "25003",
          "productName": "Dame Blanche",
          "quantity": 1.0,
          "quantityType": "PIECE",
          "unitPrice": 11.52,
          "vats": [
            {
              "label": "B",
              "price": 11.52
```

```
            }
          ]
        }
      }
    ],
    "transactionTotal": 11.52
  },
  "vatNo": "BE0000000097"
}
```


Ticket imprimé (`evidence/tickets/ROUND1_ticket.txt`, rendu texte du flux ESC/POS) :

```
            TICKET DE CAISSE T.V.A.
The FPS Finances Experience
Unite d'etablissement 2000000042
18.10.2026                                  06:03
Utilisateur: Test 12345678958
Terminal: POS-1-BAR
Ticketnr.: 000001
POS Serial: CPOS0031234567
Input device MAC: F0:1D:3D:CC:13:C8
Logiciel de caisse v.: pili-pos/1.0.0
.............................................
1 Dame Blanche                           11,52  B
TOTAL :                                   €11,52
.............................................
Mode de paiement :
Especes                                    11,52
Arrondi                                    -0,02
% TVA   CODE          HT       TVA      Brut
12,00%  B          10,29      1,23     11,52
-----------------------------------------------
               INFORMATIONS FDM
-----------------------------------------------
fdmId: MCK00000001;
fdmDateTime: 2026-10-18T04:03:55Z;
Event: N; eventCounter: 1;
totalCounter: 1; shortSignature:
9e7a9f8d01bbe357ac02a01c53f9a3b057cead56;

 [QR: https://www.gs2.be/M0000100001AbCdEfGhI]


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Experience
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TVA: BE0000000097




✂-----------------------------------------------
```

ROUND2_Arrondi_cash_Tapas_Variation

Test de base 1, situation 2. Vente directe de 1x Tapas Variation (3,34 EUR), payee uniquement en especes. Attendu : a payer 3,35 EUR, arrondi +0,01 EUR.

Preuve pili-pos (build `cert`) - situation rejouée par `crates/pos-engine/tests/spf_e2e_rounding.rs::round2_tapas_` par la commande que l'écran d'encaissement appelle. Requête **construite et archivée par pili-pos** (`evidence/uc/ROUND2_event1_N_request.json`) ; réponse FDM archivée : `evidence/uc/ROUND2_event1_N_response`


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "083b0d4a-c42c-4c72-8811-c62071a05886",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "12345678958",
  "estNo": "2000000042",
  "financials": [
    {
      "amount": 3.34,
      "amountType": "PAYMENT",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    },
    {
      "amount": 0.01,
      "amountType": "ROUNDING",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:55.006+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "transaction": {
    "transactionLines": [
      {
        "lineTotal": 3.34,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "1",
          "departmentName": "Desserts",
          "productId": "28007",
          "productName": "Tapas Variation",
          "quantity": 1.0,
          "quantityType": "PIECE",
          "unitPrice": 3.34,
          "vats": [
            {
              "label": "B",
              "price": 3.34
```

```
            }
          ]
        }
      }
    ],
    "transactionTotal": 3.34
  },
  "vatNo": "BE0000000097"
}
```


Ticket imprimé (`evidence/tickets/ROUND2_ticket.txt`, rendu texte du flux ESC/POS) :

```
            TICKET DE CAISSE T.V.A.
The FPS Finances Experience
Unite d'etablissement 2000000042
18.10.2026                                  06:03
Utilisateur: Test 12345678958
Terminal: POS-1-BAR
Ticketnr.: 000001
POS Serial: CPOS0031234567
Input device MAC: F0:1D:3D:CC:13:C8
Logiciel de caisse v.: pili-pos/1.0.0
.............................................
1 Tapas Variation                          3,34  B
TOTAL :                                     €3,34
.............................................
Mode de paiement :
Especes                                      3,34
Arrondi                                      0,01
% TVA   CODE          HT       TVA      Brut
12,00%  B           2,98      0,36      3,34
---------------------------------------------
               INFORMATIONS FDM
---------------------------------------------
fdmId: MCK00000001;
fdmDateTime: 2026-10-18T04:03:55Z;
Event: N; eventCounter: 1;
totalCounter: 1; shortSignature:
9e7a9f8d01bbe357ac02a01c53f9a3b057cead56;

 [QR: https://www.gs2.be/M0000100001AbCdEfGhI]


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✂---------------------------------------------
```

ROUND3_Arrondi_cash_vente_combinee

Test de base 1, situation 3. Vente directe de 1x Dame Blanche et 1x Tapas Variation (14,86 EUR), payee uniquement en especes. Attendu : a payer 14,85 EUR, arrondi -0,01 EUR.

Preuve pili-pos (build `cert`) - situation rejouée par `crates/pos-engine/tests/spf_e2e_rounding.rs::round3_both_c` par la commande que l'écran d'encaissement appelle. Requête **construite et archivée par pili-pos** (`evidence/uc/ROUND3_event1_N_request.json`) ; réponse FDM archivée : `evidence/uc/ROUND3_event1_N_response`


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "9984641d-825e-4b5d-bfcc-bbc1656ab3fc",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "12345678958",
  "estNo": "2000000042",
  "financials": [
    {
      "amount": 14.86,
      "amountType": "PAYMENT",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    },
    {
      "amount": -0.01,
      "amountType": "ROUNDING",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:55.007+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "transaction": {
    "transactionLines": [
      {
        "lineTotal": 11.52,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "1",
          "departmentName": "Desserts",
          "productId": "25003",
          "productName": "Dame Blanche",
          "quantity": 1.0,
          "quantityType": "PIECE",
          "unitPrice": 11.52,
          "vats": [
            {
              "label": "B",
              "price": 11.52
```

```
            }
          ]
        }
      },
      {
        "lineTotal": 3.34,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "1",
          "departmentName": "Desserts",
          "productId": "28007",
          "productName": "Tapas Variation",
          "quantity": 1.0,
          "quantityType": "PIECE",
          "unitPrice": 3.34,
          "vats": [
            {
              "label": "B",
              "price": 3.34
            }
          ]
        }
      }
    ],
    "transactionTotal": 14.86
  },
  "vatNo": "BE0000000097"
}
```


Ticket imprimé (`evidence/tickets/ROUND3_ticket.txt`, rendu texte du flux ESC/POS) :

```
            TICKET DE CAISSE T.V.A.
The FPS Finances Experience
Unite d'etablissement 2000000042
18.10.2026                                  06:03
Utilisateur: Test 12345678958
Terminal: POS-1-BAR
Ticketnr.: 000001
POS Serial: CPOS0031234567
Input device MAC: F0:1D:3D:CC:13:C8
Logiciel de caisse v.: pili-pos/1.0.0
.............................................
1 Dame Blanche                          11,52  B
1 Tapas Variation                        3,34  B
TOTAL :                                 €14,86
.............................................
Mode de paiement :
Especes                                  14,86
Arrondi                                  -0,01
% TVA   CODE          HT       TVA      Brut
12,00%  B          13,27      1,59     14,86
---------------------------------------------
               INFORMATIONS FDM
---------------------------------------------
fdmId: MCK00000001;
fdmDateTime: 2026-10-18T04:03:55Z;
```

```
Event: N; eventCounter: 1;
totalCounter: 1; shortSignature:
9e7a9f8d01bbe357ac02a01c53f9a3b057cead56;

 [QR: https://www.gs2.be/M0000100001AbCdEfGhI]


Denomination legale : The FPS Finances
Experience
Boulevard Albert II 33, 1030 Brussels
TVA: BE0000000097




✂----------------------------------------------
```

ROUND4_Cheques_repas_puis_cash

Test de base 2, situation 4. Memes produits (14,86 EUR) ; 11,52 EUR en cheques-repas (jamais arrondis), le solde 3,34 EUR en especes. Attendu : a payer 3,35 EUR en especes, arrondi +0,01 EUR.

Preuve pili-pos (build `cert`) - situation rejouée par `crates/pos-engine/tests/spf_e2e_rounding.rs::round4_meal_v` par la commande que l'écran d'encaissement appelle. Requête **construite et archivée par pili-pos** (`evidence/uc/ROUND4_event1_N_request.json`) ; réponse FDM archivée : `evidence/uc/ROUND4_event1_N_response`


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "d654884a-fd93-413d-8646-3cc0dd848489",
  "deviceId": "F0:1D:3D:CC:13:C8",
  "employeeId": "12345678958",
  "estNo": "2000000042",
  "financials": [
    {
      "amount": 11.52,
      "amountType": "PAYMENT",
      "id": "3",
      "inputMethod": "MANUAL",
      "name": "Meal voucher",
      "type": "CHEQUE_MEAL"
    },
    {
      "amount": 3.34,
      "amountType": "PAYMENT",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
    },
    {
      "amount": 0.01,
      "amountType": "ROUNDING",
      "id": "1",
      "inputMethod": "MANUAL",
      "name": "Cash",
      "type": "CASH"
```

```
    }
  ],
  "language": "FR",
  "posDateTime": "2026-10-18T06:03:55.007+02:00",
  "posFiscalTicketNo": 1,
  "posId": "CPOS0031234567",
  "posSwVersion": "pili-pos/1.0.0",
  "terminalId": "POS-1-BAR",
  "ticketMedium": "PAPER",
  "transaction": {
    "transactionLines": [
      {
        "lineTotal": 11.52,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "1",
          "departmentName": "Desserts",
          "productId": "25003",
          "productName": "Dame Blanche",
          "quantity": 1.0,
          "quantityType": "PIECE",
          "unitPrice": 11.52,
          "vats": [
            {
              "label": "B",
              "price": 11.52
            }
          ]
        }
      },
      {
        "lineTotal": 3.34,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "1",
          "departmentName": "Desserts",
          "productId": "28007",
          "productName": "Tapas Variation",
          "quantity": 1.0,
          "quantityType": "PIECE",
          "unitPrice": 3.34,
          "vats": [
            {
              "label": "B",
              "price": 3.34
            }
          ]
        }
      }
    ],
    "transactionTotal": 14.86
  },
  "vatNo": "BE0000000097"
}
```


Ticket imprimé (`evidence/tickets/ROUND4_ticket.txt`, rendu texte du flux ESC/POS) :

```
            TICKET DE CAISSE T.V.A.
The FPS Finances Experience
Unite d'etablissement 2000000042
18.10.2026                                    06:03
Utilisateur: Test 12345678958
Terminal: POS-1-BAR
Ticketnr.: 000001
POS Serial: CPOS0031234567
Input device MAC: F0:1D:3D:CC:13:C8
Logiciel de caisse v.: pili-pos/1.0.0
.............................................
1 Dame Blanche                            11,52  B
1 Tapas Variation                          3,34  B
TOTAL :                                   €14,86
.............................................
Mode de paiement :
Cheques-repas                               11,52
Especes                                      3,34
Arrondi                                      0,01
% TVA   CODE          HT       TVA      Brut
12,00%  B          13,27      1,59     14,86
---------------------------------------------
               INFORMATIONS FDM
---------------------------------------------
fdmId: MCK00000001;
fdmDateTime: 2026-10-18T04:03:55Z;
Event: N; eventCounter: 1;
totalCounter: 1; shortSignature:
9e7a9f8d01bbe357ac02a01c53f9a3b057cead56;

 [QR: https://www.gs2.be/M0000100001AbCdEfGhI]


Denomination legale : The FPS Finances
Experience
Boulevard Albert II 33, 1030 Brussels
TVA: BE0000000097




✂---------------------------------------------
```

ROUND5_Multi_paiements

Test de base 3, situation 5. Memes produits (14,86 EUR) ; 3,13 EUR en cheques-repas, 3,97 EUR par carte de debit, le solde 7,76 EUR en especes. Attendu : 7,75 EUR en especes, arrondi -0,01 EUR ; seul le paiement en especes est arrondi.

Preuve pili-pos (build `cert`) - situation rejouée par `crates/pos-engine/tests/spf_e2e_rounding.rs::round5_meal_v` par la commande que l'écran d'encaissement appelle. Requête **construite et archivée par pili-pos** (`evidence/uc/ROUND5_event1_N_request.json`) ; réponse FDM archivée : `evidence/uc/ROUND5_event1_N_response`


```
{
  "bookingDate": "2026-10-18",
  "bookingPeriodId": "34195aa5-9e4a-4968-aba4-ae982a8b47c5",
```

```
"deviceId": "F0:1D:3D:CC:13:C8",
"employeeId": "12345678958",
"estNo": "2000000042",
"financials": [
  {
    "amount": 3.13,
    "amountType": "PAYMENT",
    "id": "3",
    "inputMethod": "MANUAL",
    "name": "Meal voucher",
    "type": "CHEQUE_MEAL"
  },
  {
    "amount": 3.97,
    "amountType": "PAYMENT",
    "id": "2",
    "inputMethod": "MANUAL",
    "name": "Card",
    "type": "CARD_DEBIT"
  },
  {
    "amount": 7.76,
    "amountType": "PAYMENT",
    "id": "1",
    "inputMethod": "MANUAL",
    "name": "Cash",
    "type": "CASH"
  },
  {
    "amount": -0.01,
    "amountType": "ROUNDING",
    "id": "1",
    "inputMethod": "MANUAL",
    "name": "Cash",
    "type": "CASH"
  }
],
"language": "FR",
"posDateTime": "2026-10-18T06:03:55.008+02:00",
"posFiscalTicketNo": 1,
"posId": "CPOS0031234567",
"posSwVersion": "pili-pos/1.0.0",
"terminalId": "POS-1-BAR",
"ticketMedium": "PAPER",
"transaction": {
  "transactionLines": [
    {
      "lineTotal": 11.52,
      "lineType": "SINGLE_PRODUCT",
      "mainProduct": {
        "departmentId": "1",
        "departmentName": "Desserts",
        "productId": "25003",
        "productName": "Dame Blanche",
        "quantity": 1.0,
```

```
          "quantityType": "PIECE",
          "unitPrice": 11.52,
          "vats": [
            {
              "label": "B",
              "price": 11.52
            }
          ]
        }
      },
      {
        "lineTotal": 3.34,
        "lineType": "SINGLE_PRODUCT",
        "mainProduct": {
          "departmentId": "1",
          "departmentName": "Desserts",
          "productId": "28007",
          "productName": "Tapas Variation",
          "quantity": 1.0,
          "quantityType": "PIECE",
          "unitPrice": 3.34,
          "vats": [
            {
              "label": "B",
              "price": 3.34
            }
          ]
        }
      }
    ],
    "transactionTotal": 14.86
  },
  "vatNo": "BE0000000097"
}
```


Ticket imprimé (`evidence/tickets/ROUND5_ticket.txt`, rendu texte du flux ESC/POS) :

```
            TICKET DE CAISSE T.V.A.
The FPS Finances Experience
Unite d'etablissement 2000000042
18.10.2026                                  06:03
Utilisateur: Test 12345678958
Terminal: POS-1-BAR
Ticketnr.: 000001
POS Serial: CPOS0031234567
Input device MAC: F0:1D:3D:CC:13:C8
Logiciel de caisse v.: pili-pos/1.0.0
.............................................
1 Dame Blanche                           11,52  B
1 Tapas Variation                         3,34  B
TOTAL :                                  €14,86
.............................................
Mode de paiement :
Cheques-repas                              3,13
Carte                                      3,97
Especes                                    7,76
```

```
Arrondi                                      -0,01
% TVA   CODE          HT       TVA      Brut
12,00%  B          13,27      1,59     14,86
------------------------------------------------
                INFORMATIONS FDM
------------------------------------------------
fdmId: MCK00000001;
fdmDateTime: 2026-10-18T04:03:55Z;
Event: N; eventCounter: 1;
totalCounter: 1; shortSignature:
9e7a9f8d01bbe357ac02a01c53f9a3b057cead56;

 [QR: https://www.gs2.be/M0000100001AbCdEfGhI]


Denomination legale : The FPS Finances
Experience
Boulevard Albert II 33, 1030 Brussels
TVA: BE0000000097




✂------------------------------------------------
```

Recapitulatif

Famille	UC (spec)	UC dans le perimetre v1.0.0	Lecture des fixtures	Rejeu applicatif (build `cert`)
Basic Set v1.8	22	21 (hors M112A)	OK (22/22)	OK : 19 UC rejoues et compares a la fixture dans `spf_e2e_fixtures.rs` (M110, M111, M112B, M121, M122, M123, M130, M131, M132, M133, M140, M141, M150, M160, M161, M170, M171, M171B, M172) ; M112A n'est pas un parcours propose (remboursement partiel toujours en ticket separe, parcours M112B) ; Q100/Q110 : `client_status.rs`

Famille	UC (spec)	UC dans le perimetre v1.0.0	Lecture des fixtures	Rejeu applicatif (build `cert`)
Reports v1.3	5	5	OK (5/5)	OK : journee T1–T8 jouee puis M180, M182 et M184 compares a la fixture dans `spf_e2e_report_fixtures.rs` ; M181/M183 signes par le Z combine de M184
Advanced Set v1.6 (factures)	2	2	OK	OK : UC4121 et UC4122 rejoues dans `invoice_advanced.rs`
Total	**29**	**28**	**OK**	**OK**

Preuves d'annexe : **58 paires requete/reponse** construites et archivees par la build `cert` pour le Basic Set (M150 compris) et les rapports (`docs/SUBMISSION_ARTIFACTS/evidence/uc/`), plus un fichier `<UC>_check.json` par UC du Basic Set rejoue (comparaison champ par champ avec la fixture), **5 paires** pour les situations d'arrondi (`ROUND1` a `ROUND5`, avec leur ticket dans `evidence/tickets/`) et 10 paires pour six UC de l'Advanced Set rejoues par le meme harnais (UC223, UC225A, UC229B, UC230, UC4109, UC4119 ; voir *COUVERTURE_ADVANCED_SET_FR.md*), plus **10 paires** pour les deux factures de l'Advanced Set (UC4121 : 4 ; UC4122 : 6, dont les deux factures et le remboursement).

Pour le detail des couvertures, voir *COUVERTURE_FIXTURES_FR.md* et *COUVERTURE_RAPPORTS_FR.md*.